



Utilizing Transformer Models for Analysing Unstructured Feedback in Recommender Systems

by
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Dedicated to my family for their constant support and encouragement. To my father, for instilling the importance of education and discipline. To my mother, for always believing in my abilities and helping me see myself as capable.

Declaration

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Abstract

This study investigates how transformer-based embeddings and model architectures influence the performance of review-based recommender systems that rely on unstructured textual feedback. Recommendation Systems are increasingly using user generated reviews to better capture preferences that are not observable through structured signals alone. Textual reviews are rich in context, sentiment and product-specific detail, yet they introduce methodological challenges due to their unstructured nature, semantic complexity and noise. Advances in Natural Language Processing, particularly transformer models, offer new opportunities to model this information more effectively. As such, this study examines whether transformer-based embeddings and compact transformer backbones provide measurable improvements over traditional static embeddings and CNN-based architectures (DeepCoNN).

The research focuses on two embedding pipelines—static GloVe 6B-100d embeddings and contextual MiniLM-L6-v2 sentence embeddings and compares two downstream architectures: DeepCoNN model architecture and a lightweight transformer backbone (*prajjwal1/bert-tiny*). These models are evaluated on the Amazon All.Beauty dataset using two predictive tasks: regression-based rating prediction (RMSE and MAE) and three-class sentiment classification (Negative, Neutral, Positive) assessed through accuracy and per-class metrics.

The results show that embedding choice is the primary determinant of performance. MiniLM-L6-v2 consistently outperforms GloVe across all tasks, reflecting the importance of contextual representations for capturing nuanced and neutral sentiment in review text. The best-performing configuration, MiniLM-L6-v2 with *prajjwal1/bert-tiny*, achieves a classification

accuracy of 82.30% and the lowest MAE of 0.6673. Meanwhile, MiniLM-L6-v2 paired with DeepCoNN attains the lowest RMSE (1.0022), indicating stronger control of large errors. Architectural effects, therefore, present clear trade-offs: the transformer backbone provides more consistent predictions overall, while the CNN-based model better limits extreme deviations.

A detailed analysis of the classification results also revealed a persistent structural difficulty in modelling neutral sentiment. No configuration achieves a balanced recall across negative, neutral, and positive classes. GloVe-based models behave largely as polarity detectors, whereas MiniLM-based models improve neutral recognition but introduce new trade-offs, such as reduced negative sensitivity or over-prediction of positivity. These patterns align with previous studies reporting similar difficulties and suggest that review ratings function more like a continuous scale than a discrete sentiment taxonomy.

The study contributes a controlled evaluation of embedding and architectural choices for unstructured feedback in RS, demonstrates the viability of lightweight transformer backbones under constrained computational resources, and provides fine-grained insight into the challenges of three-class sentiment modelling. Practical implications include the prioritisation of contextual embeddings in production RS and caution when relying on discrete neutrality predictions. The findings motivate future work involving larger transformer models, extensive fine tuning of model architectures and more balanced and different sets of datasets. Full code is available at: <https://github.com/tinomuzo99/Research-Assignment>

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Nomenclature

Acronyms

AI	Artificial Intelligence
Attn	Attention
BENEFICT	BERT-Based Neural Collaborative Filtering and Fixed-Length Contiguous Tokens
BERT4Rec	Bidirectional Encoder Representations from Transformers for Sequential Recommendation
BERT	Bidirectional Encoder Representations from Transformers
CLM	Causal Language Modelling
CNN	Convolution Neural Network
CRISP-DM	Cross-Industry Standard Process for Data Mining
DeepCoNN	Deep Cooperative Neural Network
ELECTRA	Efficiently Learning an Encoder that Classifies Token Replacements Accurately
FM	Factorisation Machine
FN	False Negative
FP	False Positive

GloVe	Global Vectors for Word Representation
GPT-2	Generative Pre-trained Transformer 2
KDE	Kernel Density Estimator
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MLM	Masked Language Modeling
MPNet	MPNet
MSE	Mean Squared Error
NARRE	NARRE
NCF	Neural Collaborative Filtering
NDCG	Normalized Discounted Cumulative Gain
PLM	Permutation Language Modelling
PRM-RR	PRM-RR
RecSys	Recommender System (s)
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
RoBERTa	Robustly Optimized BERT Pretraining Approach
RS	Recommender System(s)
SSE-PT	Stochastic Shared Embeddings Personalised Transformer
TN	True Negative
TP	True Positive
Transformers4Rec	Transformers4Rec

Transformer-XL Transformer-XL

U-BERT U-BERT

word2vec word2vec

XLNet XLNet

Greek Symbols

γ Gamma

λ Mean Poisson rate

Roman Symbols

c Embedding dimension of each word vector

g Bias vector in the fully connected layer

J Output score produced by the Factorisation Machine layer

n Number of words in the input review

O Feature vector composed of all max-pooled outputs

t Filter (window) width for convolution

W Weight matrix in the fully connected layer

Subscripts

b_j Bias term for the j -th convolution filter

o_j Max-pooled feature corresponding to convolution filter j

$\hat{v}_i$ Latent factor vector associated with feature $\hat{z}_i$

$\hat{w}_0$ Global bias term in the Factorisation Machine

$\hat{w}_i$ First-order FM weight for feature $\hat{z}_i$

x_u	User latent feature representation produced by the CNN
y_i	Item latent feature representation produced by the CNN
$\hat{z}$	Concatenated user-item vector (x_u, y_i)
$\hat{z}_i$	The i -th feature of the concatenated vector $\hat{z}$
z_j	Feature map produced by applying convolution filter j
i	Subscript i refers to index of product

Other Symbols

$f(x)$	Activation function; here the Rectified Linear Unit (ReLU)
$\langle \hat{v}_i, \hat{v}_j \rangle$	Dot product between FM latent vectors $\hat{v}_i$ and $\hat{v}_j$
$V_{u_{1:n}}$	Embedded review matrix for user u across words 1 to n

Chapter 1

Introduction

1.1 Background

A Recommendation System (RS) is a category of software that employs specific algorithms to present users with suggestions for items tailored to their individual preferences (Ricci *et al.*, 2010). The term “item” refers broadly to recommended entities, which can range from consumer products to entertainment content or educational resources (Ricci *et al.*, 2010). These systems now play a central role in user decision-making for everyday digital activities, including the selection of films, music, and news articles.

Their importance lies in their extensive application in diverse domains (Zhao *et al.*, 2024). In e-commerce, platforms such as Amazon use RS to suggest products. In entertainment, services like Netflix and Spotify employ them to personalise viewing and listening experiences. Social media platforms, including TikTok and Twitter, rely on recommendation algorithms to curate content feeds, while online learning platforms such as Coursera and Udemy use them to guide learners towards relevant courses. By analysing patterns in user behaviour and preferences, RS increase user satisfaction and engagement while also contributing to business revenue.

Conventional RS are commonly classified into three primary categories: content-based methods, collaborative filtering techniques, and hybrid systems which integrate the two former approaches. Content-based systems generate recommendations by comparing other items with those the user has previously interacted with or liked, relying solely on item features to determine similarity rather than on data from other users (Ricci *et al.*, 2010). Collaborative filtering systems make recommendations by leverag-

ing the preferences of other users, thereby enabling individuals to discover items beyond those they have previously consumed (Schafer *et al.*, 2007). Hybrid systems combine these approaches, integrating content-based and collaborative methods to overcome their respective limitations. The integration of multiple techniques in hybrid models is particularly effective for mitigating issues like the "cold start" problem, a scenario characterized by a lack of historical data for new users or items. (Hasan *et al.*, 2025). However, they are often more complex and computationally expensive to implement (Su & Khoshgoftaar, 2009).

Two main forms of user input underpin RS: explicit and implicit feedback. According to Jannach *et al.* (2018), explicit feedback refers to information that users provide directly, such as numerical ratings or written reviews. In contrast, Implicit feedback derives from behavioural signals such as clicks, watch time, or purchase histories (Jannach *et al.*, 2018). Both types of feedback inform the design of different recommendation approaches.

1.1.1 Unstructured Feedback

There is a major distinction between structured and unstructured data. Structured data refers to information organised in predefined schemas, such as tabular formats or relational databases, where attributes such as ratings, demographic details, or transaction histories are stored in rows and columns and can be easily queried and analysed (Manning *et al.*, 2008). In contrast, unstructured data lack an explicit schema and cannot be readily processed by conventional database systems. It encompasses free-form sources such as emails, social media posts, survey comments, and product reviews (Sharma, 2025). According to Sharma (2025), estimates suggest that unstructured data constitute the vast majority of digital information, representing between 80% and 90% of all available data. Despite the lack of formal structure, this type of feedback provides rich contextual and sentiment analysis that structured data alone cannot capture.

Advances in natural language processing and text mining have enhanced the ability to transform unstructured feedback into actionable knowledge (Wibawa *et al.*, 2024). Such methods, as identified by Wibawa *et al.* (2024), allow researchers and practitioners to uncover latent patterns, extract sentiments, and identify themes that inform more nuanced decision-making. This is particularly relevant in domains where user perceptions and experiences play a central role.

One of the most prominent forms of unstructured feedback in the context of RS is textual review. User reviews have become a crucial source of information, complementing numerical ratings and browsing histories. Review-based approaches exploit opinionated language, sentiment expressions, product aspects, and even measures of review helpfulness (Musat *et al.*, 2013; Raghavan *et al.*, 2012; Zhang *et al.*, 2013). They capture subtle details of user preferences and item characteristics that structured signals alone cannot provide. Importantly, reviews contribute not only to the accuracy of recommendations, but also to their transparency and interpretability (Wu *et al.*, 2019).

However, the use of textual reviews also raises methodological challenges. Extracting fine-grained preferences from heterogeneous and sometimes noisy texts requires sophisticated modelling. Moreover, issues of bias, data sparsity, and privacy must be addressed (Raza *et al.*, 2024). Consequently, this study focuses on textual reviews as a source of unstructured feedback. This approach is chosen for its potential to overcome the constraints of traditional rating-based systems and to explore the capabilities of transformer-based models in RS.

1.1.2 Transformer-Based RS

The rapid growth of digital ecosystems has intensified the need for intelligent systems capable of interpreting vast volumes of unstructured textual data. RS rely on user-generated feedback such as reviews, comments, and ratings to enhance personalisation. A significant challenge lies in extracting meaningful patterns and insights from this feedback, which is characteristically noisy, context-rich, and semantically complex (Almahairi *et al.*, 2015; Baral & Li, 2016; Cheng *et al.*, 2018). In this context, the emergence of transformer models has revolutionised the field of NLP, providing a robust mechanism for modelling long-range dependencies, semantic nuances (Pota *et al.*, 2021; Riyadh & Shafiq, 2022; Tao & Fang, 2020), and structural patterns in unstructured textual data (Mienye & Swart, 2024).

The transformer model, introduced by Vaswani *et al.* (2017), represents a shift from earlier sequence models that relied primarily on recurrent or convolutional neural networks. These earlier sequence models struggled with parallelisation and effective long-distance dependency modelling due to their inherently sequential architectures (Vaswani *et al.*, 2017). In contrast, the transformer dispenses with recurrence entirely

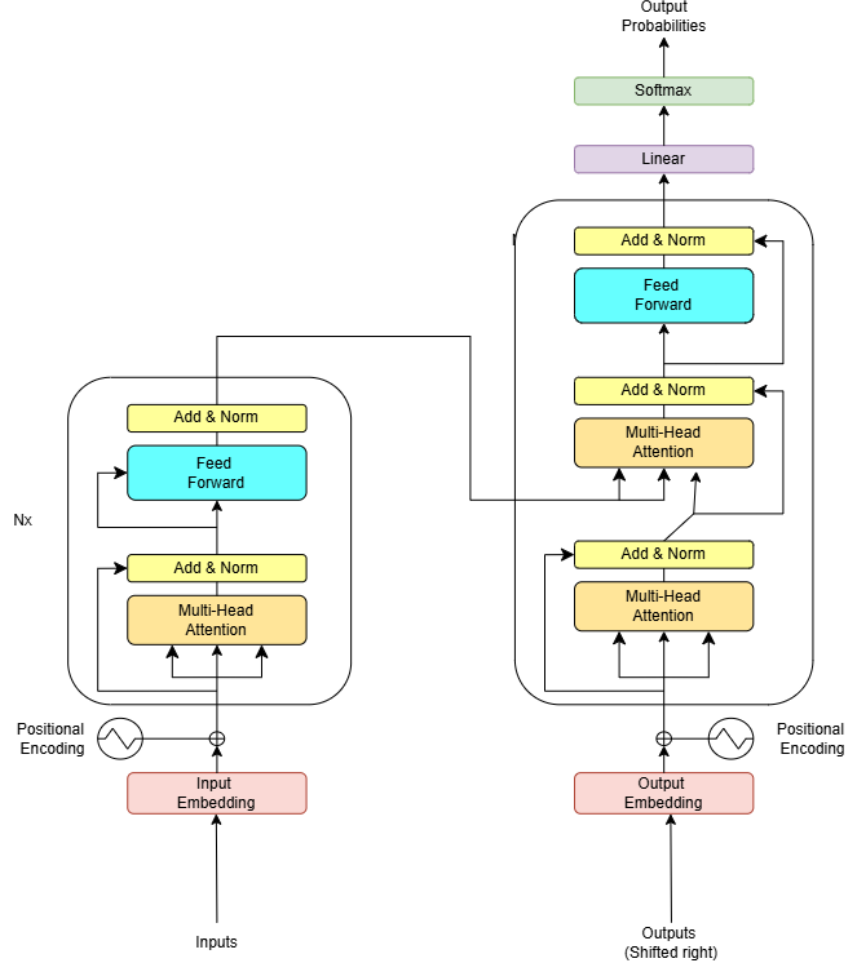


Figure 1.1: Transformer model architecture redrawn from Vaswani *et al.* (2017).

and leverages a self-attention mechanism to process input sequences holistically and efficiently.

As shown in Figure 1.1, the core of the transformer architecture is its multi-head self-attention mechanism, which enables the model to weigh the relevance of each word in a sequence relative to every other word, irrespective of positional distance. This mechanism, introduced by Vaswani *et al.* (2017), enhances the model’s capacity to capture semantic relationships while maintaining a constant path length between tokens, thereby supporting the learning of long-range dependencies, a critical capability when analysing opinion-rich customer feedback. Each attention head can learn different linguistic attributes, including co-reference, topic structure and sentiment cues, which

are necessary for interpreting the latent preferences and sentiments embedded within user-generated text.

The model’s encoder–decoder architecture, also formalised by Vaswani *et al.* (2017), is central to sequence-to-sequence tasks in which an input sequence is transformed into a meaningful output sequence. In applications relevant to recommender systems, such as review summarisation, sentiment classification or user-intent prediction, this architecture enables the system to encode textual input into high-dimensional latent representations and decode these into actionable insights or structured outputs. Positional encodings are incorporated to compensate for the absence of recurrence, allowing the transformer to retain information about word order and syntactic flow.

In the context of this research, the application of transformer models to unstructured feedback enables nuanced understanding of user behaviour and preferences beyond traditional quantitative metrics. By capturing subtle cues such as implied sentiment, sarcasm, contextual emphasis, and comparative expressions, transformer-based architectures provide a richer foundation for personalisation and user satisfaction in RS.

1.2 Research Aim, Objectives and Questions

1.2.1 Research Aim

The general aim of this study is to investigate the application of transformer models for analysing unstructured feedback in RS. Using textual reviews, the research seeks to evaluate the extent to which transformer models can enhance recommendation performance compared to traditional approaches that rely on static embeddings or simpler architectures.

1.2.2 Research Objectives

To achieve this aim, the study pursues the following objectives:

1. Implement two feature pipelines using static embeddings (GloVe 6B-100d) and contextual transformer-based embeddings (MiniLM-L6-v2).

2. Develop and compare alternative architectures, including transformer-based encoders as the backbone, a Convolutional Neural Network (CNN) baseline following a DeepCoNN-style design, and a pretrained transformer model backbone (BERT-tiny).
3. Conduct a comparative evaluation of these models using standard performance measures such as MAE and RMSE, with the potential extension to ranking-based metrics.
4. Analyse the strengths and weaknesses of each modelling approach, highlighting implications for integrating unstructured feedback in RS.

1.2.3 Research Questions

Based on these objectives, the study addresses the following research questions:

1. How do transformer-based embeddings compare to static word embeddings in representing unstructured user feedback?
2. Does a transformer-based backbone architecture outperform a CNN baseline when provided with identical input features?
3. What is the combined effect of employing transformer embeddings together with a transformer backbone on recommendation accuracy and efficiency?

1.3 Research Project Scope

This section outlines the boundaries and central focus of the study, setting out the areas that will be included to ensure a clear and manageable investigation. The scope has been designed to achieve a comprehensive yet focused exploration of the research problem, balancing depth and relevance while recognising inherent limitations. The study is guided by the overarching objective of utilising transformer-based models to analyse unstructured feedback in RS, with particular emphasis on textual reviews as a data source. To achieve this objective, the research adopts the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology. This framework offers a structured and widely recognized approach for executing data mining projects. The analysis is

limited to datasets that contain textual review information alongside traditional rating or interaction data, allowing a comparison of transformer-based architectures against alternative methods such as convolutional neural networks and embedding-based approaches. Areas outside the scope of this research include qualitative interview data and broader market-level analysis, since the study is focused on the computational modelling of review data rather than on external behavioural or sociological perspectives.

The thesis is organised into five (5) chapters to provide a logical and coherent argument. Chapter 1 introduces the research by outlining the background, relevance, and scope of the project. Chapter 2 presents the literature review, describing how the systematic review was conducted and critically analysing existing studies to identify research gaps. Chapter 3 details the research methodology, highlighting the adoption of the CRISP-DM framework and describes how it structures the phases of the study, including data understanding, preparation, modelling, and evaluation. Chapter 4 presents the results of the empirical analyses and provides a discussion of their implications in relation to the research questions and the existing literature. Finally, Chapter 5 concludes the thesis by summarising the findings, reflecting on the contributions made to both academic knowledge and practice, acknowledging limitations, and proposing directions for future work.

1.4 Contributions of the Study

This study makes several contributions to research on review-based recommender systems and the use of transformer models for unstructured feedback.

First, it provides a controlled comparison of static embeddings (GloVe 6B-100d) and contextual transformer-based embeddings (MiniLM-L6-v2), clarifying their relative effectiveness for modelling sentiment-rich reviews. Second, it evaluates alternative backbone architectures: a DeepCoNN-style CNN model architecture, and a pretrained BERT-tiny model architecture using identical inputs, thereby demonstrating how architectural choices influence rating prediction accuracy. Third, it examines the combined effect of transformer embeddings and transformer-based backbones, offering evidence on how representational and architectural capacities jointly affect performance. Finally, the study offers a transparent assessment of strengths and limitations across models

using MAE and RMSE, contributing empirical insight into when transformer-based approaches offer practical value for review-based recommendation.

Chapter 2

Literature Review

2.1 Introduction

As the first step in conducting the systematic literature review, a set of research questions aligned with the focus of this study was defined to guide and structure the review process. The research questions include:

1. How are Transformer models currently being applied to analyse unstructured user feedback in recommender systems?
2. How well do Transformer-based models perform in analysing unstructured user feedback?
3. How do different Transformer architectures compare in terms of accuracy and efficiency when applied to recommender systems?
4. What gaps exist in current research at this intersection of Transformer models and recommender systems?

Following the definition of the research questions, the next phase focused on the identification of relevant academic papers. A systematic search was performed on four major databases: Scopus, Web of Science, IEEE Explore, and ACM Digital Library. The search strategy incorporated key terms related to three core concepts, namely “transformer model”, “unstructured feedback”, and “recommender systems”. The specific queries for each database are detailed in Table 2.1. The search was limited to papers published from 2017 onward, as the seminal paper ”Attention is All You Need”

(Vaswani *et al.*, 2017) was published in late 2017, making 2017/2018 the period Transformer architectures began to gain prominence.

Initial searches in the database yielded 353 records. After the removal of 16 duplicate records and three (3) records automatically marked as ineligible by the ASReview tool, a total of 353 unique records were screened based on their titles and abstracts.

Table 2.1: Summary of database search results and selected papers.

Database	Search Query	Papers
Scopus	TITLE-ABS-KEY(("transformer model*" OR "transformer architecture*" OR BERT OR "Bidirectional Encoder Representations from Transformers" OR GPT OR "Generative Pre-trained Transformer" OR "large language model*" OR LLM* OR "attention mechanism*") AND ("unstructured feedback" OR "free-text feedback" OR "user review*" OR "natural language feedback" OR "textual feedback" OR "user comment*" OR "customer review*") AND ("recommender system*" OR "recommendation system*" OR "recommendation engine*" OR "personalized recommendation*" OR "collaborative filtering" OR "content-based filtering") AND PUBYEAR > 2017	104
Web of Science	TS=("transformer model*" OR "transformer architecture*" OR BERT OR GPT OR "large language model*" OR LLM* OR "attention mechanism*") AND TS=("unstructured feedback" OR "free-text feedback" OR "user review*" OR "natural language feedback") AND TS=("recommender system*" OR "recommendation system*" OR "recommendation engine*") AND PY=(2018-2024)	95
IEEE Xplore	("Document Title": "transformer model*" OR "Document Title": "transformer architecture*" OR "Document Title": BERT OR "Document Title": GPT) AND ("Abstract": "unstructured feedback" OR "Abstract": "user review*" OR "Abstract": "textual feedback") AND ("Abstract": "recommender system*" OR "Abstract": "recommendation system*") AND "Publication Year": 2018-2024	113
Continued on next page		

Table 2.1 – continued from previous page

Database	Search Query	Papers
ACM DL	[Title: transformer OR BERT OR GPT OR “large language model” OR LLM] AND [Abstract: “unstructured feedback” OR “free-text feedback” OR “user review”] AND [Abstract: “recommender system” OR “recommendation system”] AND [Publication Date]: (≥ 2018)	41
Total		353

The screening of these 353 records was conducted using the ASReview automated tool to efficiently manage the process. After a review of titles and abstracts, 235 records were excluded due to a clear lack of relevance to research questions (e.g., articles focused on computer vision, healthcare informatics or financial data analysis). The full text of the remaining 118 reports was sought for retrieval.

Of these, 22 reports could not be retrieved due to paywall restrictions, lack of access, or unavailable full text. Consequently, 96 reports were fully assessed for eligibility against the pre-defined inclusion criteria. Studies were required to:

- Employ pretrained transformer models.
- Use unstructured feedback in the form of textual review data.
- Utilise Amazon Review Data.
- Adopt a methodology comparable to a two-tower architecture or the DeepCoNN approach.
- Provide accessible code implementations to enable reproducibility and adaptation.
- Exclude large language model (LLM)-based conversational recommender systems.

During this full-text assessment, 88 reports were excluded. The primary reasons for exclusion were: the use of LLM-based conversational systems (n=20), the lack of use of Amazon Review Data (n=17), the use of only structured data (n=15), the use of a

non-comparable methodology (n=14), the lack of pretrained transformers (n=12), and a lack of accessible code (n=10).

This rigorous selection process resulted in 8 studies being included in the final review. Although the final set of articles did not meet every single criterion simultaneously, they met the majority and were deemed most relevant to address the research questions. The entire study selection process is summarised in the PRISMA (Page *et al.*, 2021a,b) flow diagram 2.1. These papers form the core of this analysis and are summarised in the following sections, which distinguish between empirical work and survey studies.

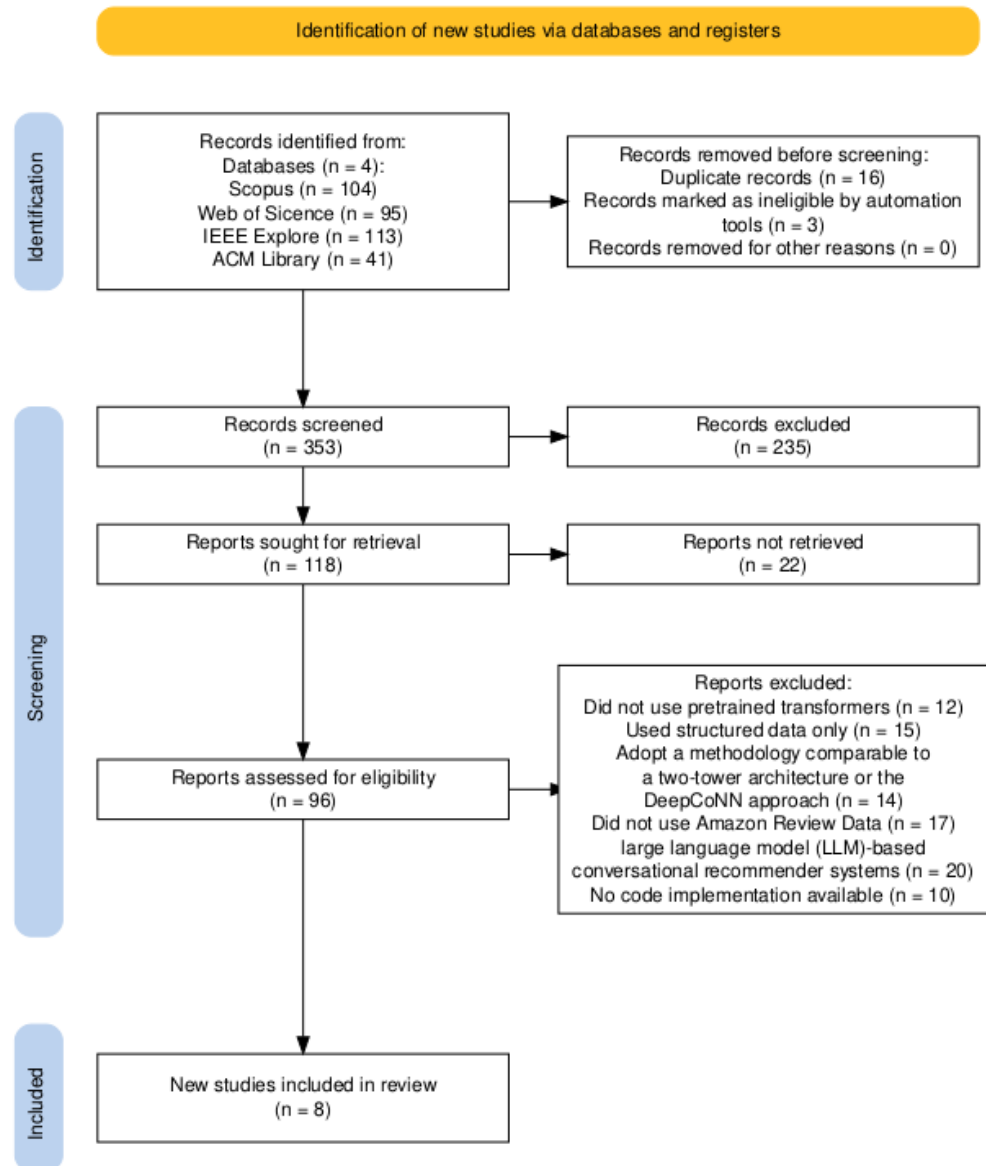


Figure 2.1: PRISMA Flow Diagram of the Study Selection Process

2.2 Empirical Studies

The empirical literature demonstrates a clear trajectory in how Transformer-based architectures have been introduced into review-based recommender systems. Early work such as DeepCoNN (Zheng *et al.*, 2017) established the CNN framework of two towers to model users and items through textual reviews. In this approach, two convolutional neural networks independently encoded user and item reviews, and the resulting representations were fused through a factorisation machine for rating prediction. DeepCoNN highlighted the potential of textual feedback to address sparsity and cold-start problems and consistently outperformed previous matrix factorisation and topic modelling approaches (Zheng *et al.*, 2017). It also became a widely used baseline, against which subsequent transformer-based methods have been benchmarked.

The introduction of transformers marked a turning point in this line of research. BERT4Rec (Sun *et al.*, 2019a) was one of the first works to adapt bidirectional transformers for sequential recommendation. It employed a Cloze-style objective, masking items within sequences and training the model to predict them using self-attention. In doing so, BERT4Rec overcame the unidirectional limitations of Recurrent Neural Network (RNN) based and CNN-based methods, achieving substantial improvements in metrics such as Hit Ratio and NDCG in multiple domains (Sun *et al.*, 2019a). The work in (Sun *et al.*, 2019a) established transformers as a strong alternative to recurrent and convolutional baselines in sequential recommendation tasks.

A further comparative perspective was offered by Gheewala *et al.* (2024), who systematically evaluated BERT, RoBERTa, and XLNet against CNN and RNN architectures on Amazon review datasets. Their findings confirmed that contextualised embeddings derived from transformers consistently outperformed static word embeddings such as word2vec (Gheewala *et al.*, 2024). They also demonstrated that fine-tuning transformer backbones yielded superior predictive accuracy, reinforcing the importance of contextual information in modelling review data. Together, Sun *et al.* (2019a) and Gheewala *et al.* (2024) provide strong evidence that Transformer-based models can enhance both rating prediction and ranking performance in review-based recommender systems.

The momentum created by these studies encouraged the development of frameworks that explored the versatility of transformers in various tasks. Transformers4Rec

(de Souza Pereira Moreira *et al.*, 2021) introduced a unified library to apply Transformer-based models to sequential and session-based recommendations. It evaluated a range of transformer backbones, including BERT, GPT-2, XLNet, and Transformer-XL, under different training conditions such as causal language modelling, masked language modelling, and replaced token detection. The results showed that the transformers not only outperformed traditional baselines but also achieved notable gains when side information and auxiliary tasks were incorporated. This work demonstrated the adaptability of transformers in diverse domains, from e-commerce to news recommendation.

Alongside these frameworks, hybrid approaches have sought to integrate transformers with collaborative filtering methods. Wang *et al.* (2025) proposed PRM-RR, which combined rating matrix factorisation with ALBERT-based textual encodings and Bi-GRU attention mechanisms. Tested on several Amazon datasets, the model showed significant gains over matrix factorisation and earlier text-enhanced recommendation methods. Similarly, BENEFICT (Pugoy & Kao, 2020) incorporated a BERT encoder within a neural collaborative filtering framework, while also introducing an explanation module based on multi-span prediction. Beyond accuracy improvements, BENEFICT emphasised interpretability, demonstrating that transformer-based representations could provide more transparent explanations of recommendations compared to CNN baselines such as DeepCoNN and NARRE (Pugoy & Kao, 2020).

Another important development has been the use of pre-training to enhance recommendation performance under sparse data conditions. Qiu *et al.* (2021) proposed U-BERT, which introduces a cross-domain pre-training strategy for learning user representations. This strategy employs pre-training tasks such as masked opinion token prediction and opinion rating prediction prior to fine-tuning on downstream recommendation tasks. This approach substantially improved performance across multiple Amazon domains and Yelp data, showing that transformer-based pre-training can alleviate sparsity and improve transferability.

Collectively, these empirical studies indicate that transformer-based methods consistently outperform CNN and RNN baselines in tasks involving textual review data. They also demonstrate additional advantages, such as improved interpretability, effective cross-domain transfer, and enhanced performance under sparse conditions. At the same time, studies reveal meaningful variation in how transformer models are applied, from sequential architectures such as BERT4Rec, to hybrid models such as PRM-RR

and BENEFICT, to pretraining strategies exemplified by U-BERT. This body of work provides the foundation against which the present study positions its comparative evaluation of static embeddings, CNN backbones, and transformer-based architectures using Amazon review data.

2.3 Survey Studies

In addition to empirical contributions, survey-based studies provide an overview of the broader landscape and help contextualise the rapid rise of transformer-based recommender systems. These reviews highlight both the progress made and the gaps that remain, offering valuable insights into how the present research aligns with ongoing trends. One such review is provided by Hasan *et al.* (2025), who surveyed over 100 studies in review-based recommender systems published between 2010 and 2021. Their work categorised existing approaches into traditional machine learning methods, CNN- and RNN-based architectures, hybrid models, and attention-based frameworks. While the survey did not empirically evaluate the methods, it systematically mapped the evolution of review-based recommendation and identified key challenges such as interpretability, sparsity, and the need for multimodal extensions. Notably, Hasan *et al.* (2025) identified transformer-based architectures as a promising but underexplored direction, positioning them as a “future perspective” rather than an established approach at the time of their review. This observation underscores the emerging role of transformers in the recommender systems domain and situates the current study within a research area that was only beginning to attract substantial attention.

A more targeted review was undertaken by Pohan *et al.* (2022), who conducted a systematic literature review of Transformer-based recommender systems published between 2019 and 2021. Using Scopus as their primary database, they applied strict inclusion and exclusion filters (English language, Transformer-related terms, recommender system focus) to identify 84 publications. Their review combined bibliometric and content analysis, mapping the distribution of studies by year, source, country, and subject area, while also highlighting the most cited transformer-based approaches, including BERT4Rec (Sun *et al.*, 2019a), SSE-PT (Wu *et al.*, 2020), and Personalised Reranking for Recommendation variant Pei *et al.* (2019). Pohan *et al.* (2022) documented rapid growth during this short period of time, noting that transformers were

increasingly recognised for their ability to capture contextual and sequential information. However, they also emphasised ongoing challenges around scalability, efficiency, and generalisability.

Together, these survey papers reinforce the trajectory established by the empirical studies. Hasan *et al.* (2025) provide a longer-term view of review-based recommendation and highlight transformers as an anticipated but underdeveloped line of research, while Pohan *et al.* (2022) capture the acceleration of transformer adoption during its initial years. Both reviews point to the same set of enduring challenges: limited interpretability, scalability constraints, and the need to extend transformer-based models beyond single modalities or datasets. These insights situate the present study within a fast-growing but still maturing field, justifying the need for focused comparative evaluations of transformer-based approaches against established baselines.

2.4 Research Gaps

Despite the demonstrated advantages of transformer-based methods, several limitations remain in the literature. Although empirical studies such as Gheewala *et al.* (2024) have shown that contextual transformer embeddings outperform static word embeddings, few works provide a direct, controlled comparison between static embeddings, CNN baselines, and transformer backbones under identical experimental conditions. This makes it difficult to isolate the specific contributions of embeddings versus architectures. Furthermore, although approaches like U-BERT (Qiu *et al.*, 2021) have explored cross-domain pre-training to mitigate sparsity, and hybrid designs such as Wang *et al.* (2025) demonstrate the potential of combining transformers with collaborative filtering, the trade-off’s between accuracy and computational efficiency remain insufficiently examined.

Survey studies reinforce these observations. Hasan *et al.* (2025) emphasised that review-based recommender systems still face unresolved challenges of interpretability and multimodal integration, while Pohan *et al.* (2022) highlighted scalability and efficiency as key barriers to wider adoption of transformer-based models. Taken together, the empirical and survey evidence points to a gap in systematic, comparative evaluations that can disentangle the relative benefits of embeddings and architectures, while also shedding light on efficiency and interpretability concerns. Addressing this gap, the

present research evaluates static and transformer-based embeddings, CNN and transformer backbones, and hybrid approaches using Amazon review data. In doing so, it aims to clarify the role of transformer models in advancing review-based recommender systems and to provide practical insights into their trade-offs in accuracy, efficiency, and interpretability.

2.5 Transformer Models in Recommender Systems

Having identified the limitations of earlier approaches to modelling user feedback, this section introduces transformer architectures as a foundational concept for the proposed study. The aim is to clarify how transformer-based models process unstructured textual input, outline the major architectural variants, and evaluate their strengths and limitations when adapted for recommendation tasks.

2.5.1 Origins and Core Properties of Transformer Architectures

With the Transformer architecture, [Vaswani *et al.* \(2017\)](#) instigated a fundamental transition in sequence modeling, displacing RNNs as the dominant approach. Transformers replace recurrence with a self-attention mechanism capable of capturing long-range dependencies, allowing parallel computation and improving scalability for long sequences. These properties established transformers as the dominant choice for numerous NLP tasks, leading to the development of influential variants such as GPT-2 ([Radford *et al.*, 2018](#)), BERT ([Devlin *et al.*, 2019](#)) and XLNet ([Yang *et al.*, 2019](#)). Although conceptually similar, these models diverge in their pre-training objectives, including causal language modelling (CLM), masked language modelling (MLM) and permutation language modelling (PLM), each offering distinct advantages for text representation learning.

2.5.2 Transformers for Sequential Recommendation

Insights from large-scale NLP models suggested that transformers were effective when trained on long sequences ([Liu *et al.*, 2019](#)), motivating their adaptation for sequential recommendation. Early work such as AttRec ([Zhang *et al.*, 2019](#)) used self-attention to model item–item dependencies within user interaction histories and incorporated metric learning to capture long-term preferences. SASRec ([Kang & McAuley, 2018](#))

adopted a GPT-style CLM objective to predict the next item using only past interactions. BERT4Rec (Sun *et al.*, 2019a) demonstrated improved accuracy using an MLM objective during training, masking items within the sequence and predicting them based on contextual information. To mitigate potential leakage of future information during inference, only the final item is masked at prediction time.

Subsequent work investigated personalisation within transformer structures. SSEPT (Wu *et al.*, 2020) augmented item embeddings with user embeddings while applying stochastic shared embeddings to reduce overfitting. These models were tested on datasets with long interaction sequences (average lengths ranging from approximately 9 to 1655), highlighting the need for architectures that can manage long-term temporal dependencies.

Temporal irregularity posed an additional challenge. Unlike word sequences, user interactions are not evenly spaced in time. Attempts to learn continuous time representations proved unstable (Zhou *et al.*, 2018). As a result, elapsed times were discretised at a log scale and encoded as categorical features. Industrial applications further emphasised the value of incorporating side information, including item metadata and contextual features (Chen *et al.*, 2019a; Lv *et al.*, 2019; Zhou *et al.*, 2018). Methods such as ATRank (Zhou *et al.*, 2018), BST (Chen *et al.*, 2019a) and SDM (Lv *et al.*, 2019) illustrated how self-attention could be used to fuse heterogeneous user behaviour signals.

2.5.3 Transformers for Session-based Recommendation

Transformer models have also been applied to session-based recommendation, where interaction sequences are shorter but require fine-grained modelling of user intent. A range of studies (Chen *et al.*, 2019b; Fang, 2021; Luo *et al.*, 2020; Pan *et al.*, 2020; Sun *et al.*, 2019b; Xiao *et al.*, 2017; Zhang *et al.*, 2020) found that transformers outperform recurrent architectures even under these shorter-sequence conditions. Some works addressed noise within sessions: preference-aware masking (Zhang *et al.*, 2020) and modified self-attention weights (Pan *et al.*, 2020) were proposed to distinguish relevant from irrelevant items. Other approaches integrated graph neural networks to capture structural dependencies (Xiao *et al.*, 2017).

Training strategies also varied. Most models adopted auto-regressive training, using only past interactions. In contrast, BERT4SessRec (Chen *et al.*, 2019b) used an

MLM-style approach, enabling the model to exploit in-session future interactions during training while still producing valid next-item predictions.

2.5.4 Transformers4Rec: A Consolidated Framework

Transformers4Rec (de Souza Pereira Moreira *et al.*, 2021) was introduced to bridge the gap between NLP-focused transformer libraries and the specific requirements of sequential and session-based recommendation. Built on the HuggingFace Transformers library, it provides a flexible interface for experimenting with different architectural variants, training objectives and feature modalities.

A comprehensive empirical analysis demonstrated that certain transformer architectures consistently outperformed strong baselines across e-commerce datasets, while showing comparable performance in news domains (de Souza Pereira Moreira *et al.*, 2021). Further experiments isolated the effects of specific training objectives for XLNet, finding that replacement token detection delivered robust performance across dataset types. The study also established that incorporating item metadata, user context features and other side information resulted in consistent performance gains.

2.6 Literature Review Summary

This section summarises the key insights from the reviewed literature in relation to the four research questions posed in Section ??.

2.6.1 Application of Transformer Models to Unstructured Feedback

The literature shows that transformer architectures have been applied to both sequential and review-based recommendation tasks. Early adaptations, such as BERT4Rec and SASRec, focused on modelling item sequences, while later developments integrated transformers directly into review-based pipelines through contextual embeddings and hybrid architectures. These models typically replace CNN- or RNN-based encoders and employ self-attention to capture semantic dependencies within user-generated text. Frameworks such as Transformers4Rec further illustrate how transformer variants can be adapted to multiple recommendation scenarios.

2.6.2 Performance of Transformer-Based Models

Across empirical studies, transformer-based models demonstrate consistent performance gains over traditional baselines when analysing unstructured textual feedback. Contextual embedding models, including BERT, RoBERTa and XLNet, outperform static word embeddings by capturing sentiment nuance and contextual meaning. These advances translate into improvements in rating prediction, ranking accuracy and robustness under sparse data conditions. Pre-training strategies, such as those employed in U-BERT, further enhance performance through cross-domain knowledge transfer.

2.6.3 Comparison of Transformer Architectures in Accuracy and Efficiency

Comparative evaluations show that different transformer architectures vary in their trade-offs between accuracy and computational cost. Bidirectional models trained with masked language modelling objectives generally achieve strong predictive performance, whereas autoregressive models offer efficiency advantages in sequential prediction. Studies also indicate that architectural choices—such as embedding dimensionality, number of layers and incorporation of user/item side information—significantly influence performance. However, evidence remains limited regarding systematic comparisons under controlled experimental conditions, particularly with respect to computational efficiency.

2.6.4 Remaining Gaps in Current Research

Four key gaps emerge across the literature. First, few studies offer controlled comparisons between static and contextual embeddings within identical modelling frameworks, making it difficult to isolate their individual contributions. Second, comparative analyses of transformer backbones remain fragmented, with limited attention to efficiency and scalability. Third, despite advances in hybrid models, interpretability remains underdeveloped. Finally, research continues to rely heavily on domain-specific datasets, with limited exploration of cross-dataset generalisability. These gaps justify the present study’s focus on evaluating embeddings and architectures under consistent conditions using Amazon review data.

Chapter 3

Methodology

3.1 Introduction

This section outlines the methodology employed in this study, which investigates the application of transformer-based models to analyse unstructured feedback within the context of recommender systems. The methodological approach is structured according to the CRISP-DM framework. This established model ensures a systematic, industry-aligned process that spans from problem understanding to final deployment. The proposed methodology builds upon the Deep Cooperative Neural Networks (DeepCoNN) model proposed by [Zheng *et al.* \(2017\)](#), adapting to optimise recommendation performance using unstructured feedback.

3.2 CRISP-DM Framework overview

The CRISP-DM framework, as illustrated in [Figure 3.1](#), consists of six iterative phases:

1. Business understanding
2. Data understanding
3. Data preparation
4. Modelling
5. Evaluation
6. Deployment

3.2 CRISP-DM Framework overview

These phases were sequentially implemented with tailored adjustments to accommodate the nuances of unstructured feedback and transformer-based modelling in recommender systems.

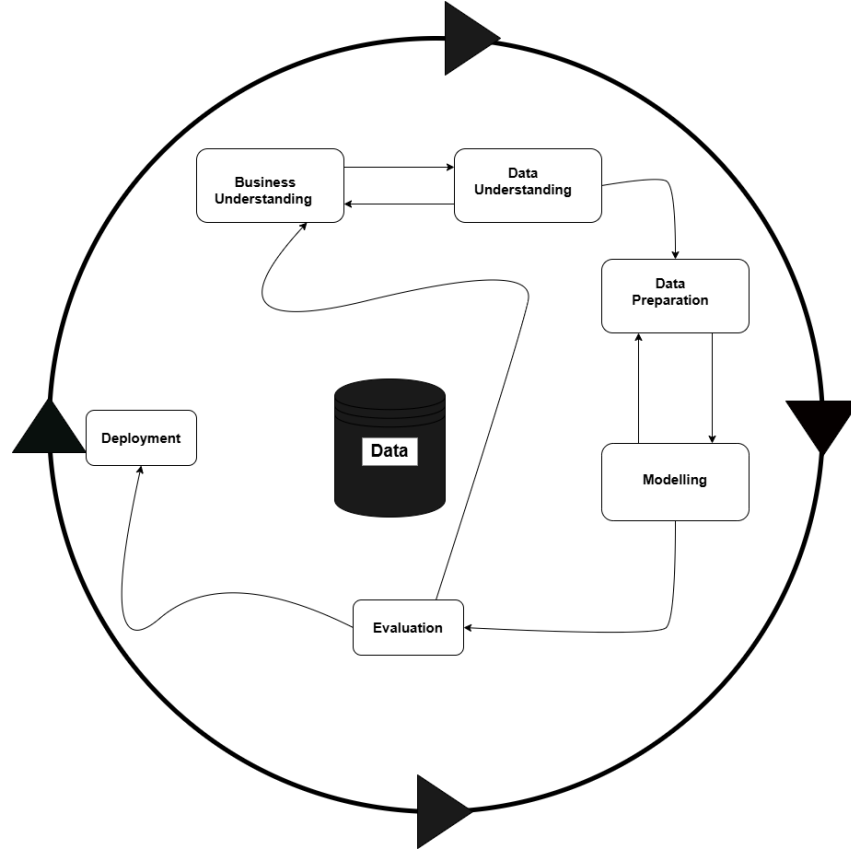


Figure 3.1: CRISP-DM Framework redrawn from [Martínez-Plumed *et al.* \(2021\)](#)

Within this framework, the Business Understanding phase involved defining the research objective of improving recommender accuracy through the analysis of unstructured textual feedback. The Data Understanding phase entailed exploring datasets that contained both rating information and textual reviews. In the Data Preparation phase, reviews were cleaned and transformed into numerical vectors using pre-trained embedding models. The Modelling phase focused on the construction of alternative deep neural architectures for user-item rating prediction. The Evaluation phase centred on assessing predictive accuracy using established metrics. Finally, the Deployment phase was considered in terms of the practical applicability and scalability of the models in industrial recommender systems.

3.2.1 Business Understanding

The central aim of this research is to enhance the quality and personalisation of recommendations by leveraging transformer models to analyse unstructured textual feedback, such as customer (user) reviews. Traditional recommender systems often rely on structured user-item ratings, which suffer from sparsity and cold-start issues (Huai *et al.*, 2023; Pan *et al.*, 2023). By employing transformer models, this study aims to extract latent user preferences and item characteristics from review text to augment existing collaborative filtering methods.

3.2.2 Data Understanding

3.2.2.1 Dataset Selection and Justification

In the domain of recommender systems, the choice of dataset plays a pivotal role in both the development and evaluation of the model’s performance. Since this research focuses on utilising transformer-based models for analysing unstructured feedback, i.e., textual reviews, the dataset must contain rich textual data along with user-item interactions and corresponding ratings Hasan *et al.* (2025). This section outlines the criteria considered for dataset selection and provides detailed justification for the chosen datasets.

3.2.2.2 Dataset Selection Criteria

To effectively train and evaluate transformer models for recommendation tasks using unstructured feedback, the following criteria were established for selecting datasets:

- **Presence of Textual Reviews:** As the primary objective is to leverage unstructured user feedback, datasets must contain a substantial number of textual reviews along with traditional rating data.
- **Scale and Diversity:** To ensure generalisability and robustness, datasets with many users, items, and reviews are preferred. In publicly available datasets, the minimum number of users is 1000 Kanwal *et al.* (2021). Diversity in domains (e.g., e-commerce, hospitality, food, etc.) is also considered to evaluate model adaptability.

- **Availability of Explicit Feedback:** While implicit feedback can be useful, the focus of this research is on datasets that include explicit feedback (e.g., ratings) in conjunction with text reviews.
- **Public Accessibility:** Only publicly available datasets are considered to ensure reproducibility of the experiments and comparability with existing research.
- **Prevalence in Literature:** Datasets that have been frequently used in recent studies are prioritised for benchmarking and aligning with state-of-the-art methodologies.
- **Sparsity and Density Information:** The sparsity level of user-item interactions is considered, as it affects model training and evaluation. According to [Gheewala *et al.* \(2024\)](#), sparsity density is calculated using the formula:

$$\text{Sparsity Density} = \frac{\text{Number of Ratings}}{\text{Number of Users} \times \text{Number of Products}}. \quad (3.1)$$

Datasets with a moderate to high number of interactions are ideal.

3.2.2.3 Selected Dataset

Based on the above criteria and insights from existing studies ([Gheewala *et al.*, 2024](#); [Hasan *et al.*, 2025](#); [Kanwal *et al.*, 2021](#)), the dataset selected for this study is the large-scale Amazon Reviews dataset collected in by [Hou *et al.* \(2024\)](#). This dataset contains:

- 48.19 million items
- 571.54 million reviews
- 54.51 million users
- Data spanning from May 1996 to September 2023

This corpus includes paired structured (ratings) and unstructured (review text) feedback, providing a rich foundation for modelling.

3.2.2.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to gain an initial understanding of the *All_Beauty* subset of the Amazon Reviews 2023 dataset by Hou *et al.* (2024). This dataset contains 701,528 customer reviews with ten main features: `rating`, `title`, `text`, `images`, `asin`, `parent_asin`, `user_id`, `timestamp`, `helpful_vote`, and `verified_purchase`. Each record represents a unique user review, including both textual and numerical data. The `rating` feature, stored as a float64 type, provides the overall product score on a five-point scale, while `title` and `text`, both string objects, contain the review headline and detailed user feedback, respectively. Additional attributes such as `asin` and `parent_asin` identify product instances, and `user_id`, also of string type, uniquely represents individual reviewers. The `timestamp` records the time of review submission, and `verified_purchase` indicates whether the reviewer had a confirmed purchase, providing a measure of data reliability.

Target Variable

The distribution of the target variable, `rating`, was then examined to understand customer feedback patterns. Frequency counts showed that ratings were highly imbalanced: 420,726 reviews (approximately 60%) were five-star, followed by 102,080 one-star reviews, 79,381 four-star reviews, 56,307 three-star reviews, and 43,034 two-star reviews. This distribution, visualised in Figure 3.2, revealed a strong right-skew towards positive evaluations.

The predominance of high ratings reflects a common trend in online consumer data, where customers are more likely to post feedback when highly satisfied. This positive bias can introduce challenges for model training, as imbalanced target distributions often reduce predictive performance for lower-frequency classes.

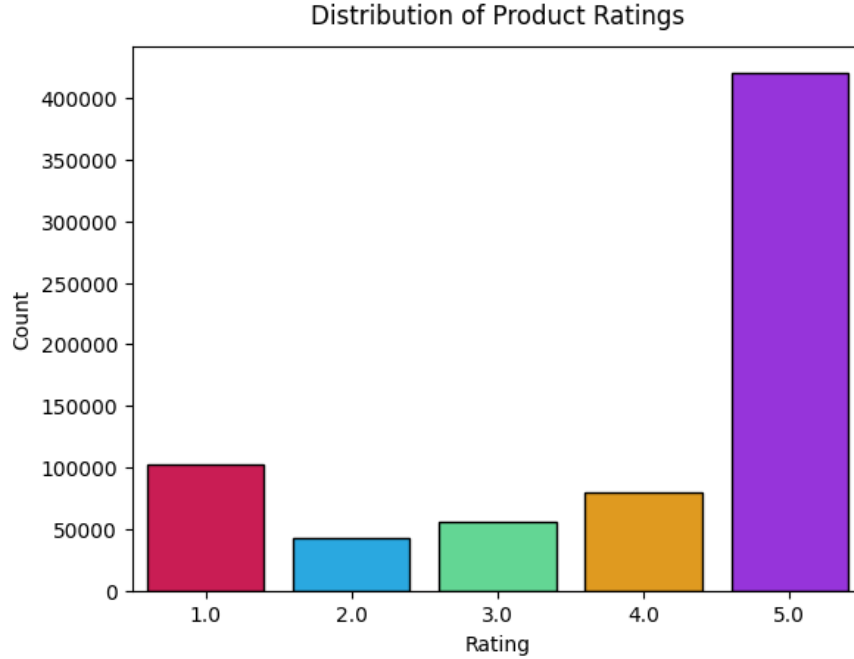


Figure 3.2: Distribution of product ratings for the *All_Beauty* dataset.

Word Cloud Analysis

Analysis of the word clouds generated for each rating category provides a clear understanding of customer sentiment differentiation across the rating spectrum. The lexical patterns confirm the visual validation of the numerical distribution of ratings, highlighting linguistic cues associated with varying levels of satisfaction and dissatisfaction.

Five-Star Reviews

The word cloud for five-star reviews revealed a strongly positive lexical pattern, characterised by frequent expressions of approval, satisfaction, and enthusiasm. The most dominant terms, such as *love*, *great*, *good*, *perfect*, *easy*, and *nice*, reflected consistently high levels of customer satisfaction and product endorsement. The prominence of affective words such as *love*, *amazing*, and *happy* suggests that customers felt their expectations had been fully met or exceeded. Reviewers frequently commented on specific product attributes, including comfort, texture, and appearance, using descriptive

3.2 CRISP-DM Framework overview

characters and a mean of approximately 159.7, while five-star reviews were shorter, with a median of 91 and a mean of 160.8. Two-, three-, and four-star reviews were slightly longer, with median values between 124 and 130 characters. For token-based lengths, one-star reviews had a median of 20 tokens, compared to 17 for five-star reviews. Intermediate ratings (two to four stars) demonstrated slightly longer median lengths ranging from 24 to 25 tokens. The 90th percentile values confirmed that most reviews were relatively concise, rarely exceeding 130 characters or 25 tokens.

Table 3.1: Summary statistics of review lengths (token count) by rating for the *All_Beauty* dataset.

Rating	Median	Mean	90th Percentile
1-star	20.0	30.27	20.0
2-star	24.0	36.75	24.0
3-star	25.0	39.00	25.0
4-star	25.0	42.55	25.0
5-star	17.0	30.26	17.0

Table 3.2: Summary statistics of review lengths (character count) by rating for the *All_Beauty* dataset.

Rating	Median	Mean	90th Percentile
1-star	105.0	159.68	105.0
2-star	124.0	192.41	124.0
3-star	127.0	203.45	127.0
4-star	130.0	223.17	130.0
5-star	91.0	160.76	91.0

The density plot, Figure 3.8, showed that review lengths were right-skewed across all ratings, with most reviews clustering at shorter lengths below 100 tokens. This pattern suggests that customers typically write brief comments, regardless of sentiment. However, reviews associated with lower ratings exhibited a broader distribution and higher variability, indicating that dissatisfied users were more likely to elaborate on negative experiences.

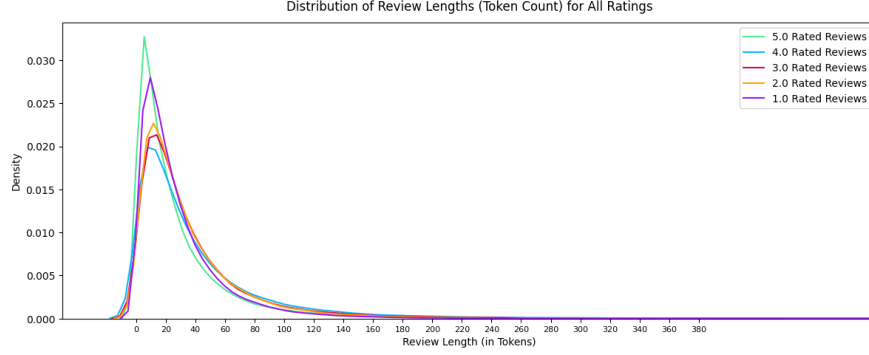


Figure 3.8: Distribution of review lengths (token count) for all rating categories in the *All_Beauty* dataset.

The box plots (Figure 3.9) across different star ratings reveals several notable patterns. Median review lengths appear broadly consistent among one- to four-star ratings, with four-star reviews tending to be slightly longer on average. In contrast, five-star reviews are generally shorter, as indicated by a lower median value. The variability in review length, represented by the height of each box, is greatest for four-star ratings, suggesting that reviewers awarding four stars exhibit more inconsistency in the amount of detail provided, as some write succinct comments while others produce considerably longer evaluations. Both one-star and five-star reviews tend to be shorter overall, which aligns with common behavioural tendencies in user feedback: dissatisfied reviewers often express their discontent briefly, whereas highly satisfied users typically provide concise praise. This pattern, characterised by moderate-length reviews at the extremes and longer, more detailed responses in the middle range, is frequently observed in consumer review data.

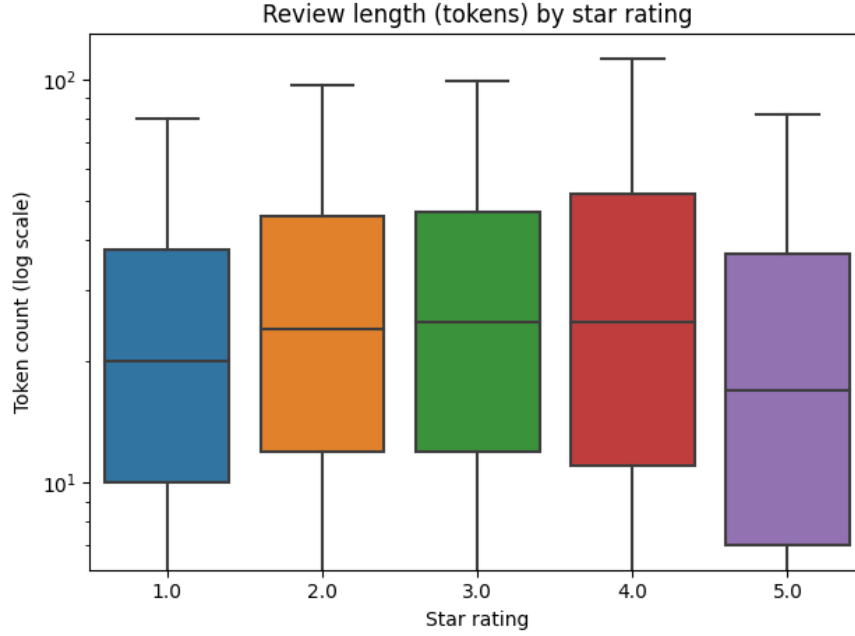


Figure 3.9: Box plot of review length (in tokens) by star rating on a logarithmic scale for the *AllBeauty* dataset.

Exploratory data analysis was conducted to assess:

- The distribution of review lengths
- Vocabulary diversity
- Frequency of rating scores
- Dataset sparsity

This analysis was critical in understanding the data landscape, influencing both preprocessing strategies and model hyperparameter tuning decisions.

3.2.3 Data preparation

3.2.3.1 Data Preprocessing

Data loading

As mentioned earlier, The *AllBeauty* subset of the Amazon Reviews dataset was used as the main data source for this research. The data were provided locally as

a JSON Lines file and were ingested using a custom loader function. This loader employed a two-stage process to ensure both efficiency and reliability. Initially, it attempted to load the file using the *pandas.read_json* function, which provides fast and direct access to JSONL data. If this approach failed, the function reverted to the Hugging Face datasets library to perform the same operation more robustly. This dual strategy ensured that the data could be accessed consistently, even when variations in file structure were present.

To achieve a uniform schema across the dataset, all fields were standardised to four standard columns: `userID`, `itemID`, `review`, and `rating`. The loader automatically mapped common alternative field names, such as `reviewText` to `text`, `overall` to `rating`, `user` to `user_id`, and `item` to `asin`. This step resolved inconsistencies that often occur in Amazon product datasets. Records containing non-numeric ratings or blank review texts were removed to maintain data integrity. Each dataset was then re-indexed to ensure sequential and contiguous entries, producing a clean and consistent `DataFrame` for further processing.

Identifiers were re-mapped to contiguous integer values using categorical encoding in *pandas*. This transformation was applied to both `userID` and `itemID` columns, resulting in dense integer ranges that improved memory efficiency and computational performance in later stages of analysis. The mapping process was deterministic within each execution, ensuring reproducibility across experiments.

Data Cleaning

Text preprocessing involved several systematic cleaning steps. All review text was converted to lowercase to eliminate case sensitivity. A curated list of punctuation symbols was removed by replacing each with a space, followed by tokenisation using the `nltk.WordPunctTokenizer`, which effectively handled a range of alphanumeric tokens. Tokens corresponding to common stopwords were removed using a predefined list. The remaining tokens were then joined to form the cleaned review text. This cleaning procedure reduced textual noise while preserving meaningful surface forms and domain-specific expressions. Importantly, stemming and lemmatisation were not applied, as they can sometimes distort product-specific terminology.

Following the data cleaning process, the dataset was partitioned into training, validation, and test subsets to support the subsequent phases of model development and

evaluation. The split was carried out using the `train_test_split` function from scikit-learn, with a training ratio of 0.8. The remaining 20 percent of data were split equally into validation and test sets. Random seeds were fixed during both split operations to ensure reproducibility and consistency of results across different runs. This process avoided data leakage between the sets, as the splits were generated after text cleaning and identifier re-indexing.

After preprocessing, the dataset contained a total of 700,808 reviews, corresponding to 631,352 unique users and 115,637 unique items. The resulting split sizes were 560,646 reviews for training, 70,081 for validation, and 70,081 for testing. These statistics highlight the high ratio of unique users and items relative to the total number of interactions, which reflects the sparse nature of e-commerce datasets and presents potential challenges for recommendation models.

Throughout the preprocessing pipeline, measures were implemented to ensure reproducibility and fault tolerance. Fixed random seeds were applied to all split operations, and comprehensive error handling was built into the data loader. The loader provided detailed diagnostic messages if both `pandas` and `datasets` failed to load the data. Similarly, the existence of external stopword and punctuation lists was verified at runtime, with informative error messages prompting corrective action if any required files were missing.

3.2.3.2 Embeddings Generation

As mentioned earlier, this methodology builds upon the DeepCoNN model. Within this framework, a user profile is constructed by concatenating all reviews authored by that user into a single, composite document. Similarly, all reviews written for a particular item are combined to form the item profile. These aggregated documents are then encoded into dense vector representations which serve as the input to parallel neural networks for users and items. Two distinct approaches to representing review text were compared. The first approach employed pre-trained word embeddings using Stanford’s Global Vectors (GloVe) with 100 dimensions (Pennington *et al.* (2014a)). This method provides a static semantic representation where each word is mapped to a fixed vector irrespective of its context. The second approach employed the all-MiniLM-L6-v2 model from the Sentence-BERT family of Transformer encoders. In contrast to GloVe, which produces static word embeddings, this model generates contextualized representations

where the meaning of each token is dynamically conditioned on its surrounding textual context. Comparing these two approaches allows for an assessment of the contribution of contextual meaning to rating prediction.

Data preparation involved multiple sub-tasks to ensure compatibility with transformer-based architectures:

- Tokenisation: Use of SentencePiece tokenizer compatible with transformer models
- Padding and Truncation: Ensured uniform sequence lengths for batch processing
- Matrix Construction: Generation of user-item rating matrices and text-review corpora

Transformer-Based Sentence Embeddings

For sentence-level embedding generation, the all-MiniLM-L6-v2 model was utilized. It functions by mapping input text to a 384-dimensional dense vector space, wherein the cosine distance between vectors reflects their semantic similarity (Wang *et al.*, 2020). It is based on nreimers/MiniLM-L6-H384-uncased and trained on more than one billion sentence pairs, making it highly suitable for general-purpose semantic representation tasks such as clustering and similarity search.

The model uses mean pooling to aggregate token-level representations from the final hidden state of the transformer layers, resulting in normalised embeddings of fixed length (384 dimensions). It supports similarity metrics including cosine similarity, dot product, and Euclidean distance, enabling flexibility for downstream modelling.

The embedding process was implemented through a custom function which pre-computed embeddings across the training, validation, and test datasets. Each dataset consisted of user-item review pairs and corresponding ratings. Texts were lowercased, tokenised, and truncated to a maximum length of 128 tokens. Each review was segmented into a fixed number of sentences ($S = 40$), and each sentence was encoded independently in batches of 1,024 using GPU acceleration.

A custom mean pooling function was applied to derive sentence-level embeddings by averaging the last hidden states, weighted by the attention mask. The resulting embeddings were stored in a NumPy memory-mapped file, with accompanying metadata files recording sentence identifiers and review-to-sentence mappings.

The all-MiniLM-L6-v2 model was selected over alternatives such as DistilBERT and BERT-base because it achieves comparable semantic performance with significantly lower computational cost. MiniLM models retain most of BERT’s representational power while being up to 60% smaller and several times faster, which is particularly advantageous for large-scale text corpora. This efficiency, combined with robust performance across semantic similarity and retrieval benchmarks, made the model well suited for this study’s requirements of scalability, accuracy, and interpretability.

Static Word Embeddings: GloVe

GloVe embeddings were computed to assess the performance of non-contextual, word-level representations. The GloVe algorithm (Pennington *et al.*, 2014b) learns word embeddings in an unsupervised manner by performing matrix factorization on the global word–word co-occurrence matrix constructed from a corpus. The method is grounded in the distributional hypothesis, which posits that words with similar contextual distributions share semantic meaning. These captured relationships are represented as linear substructures in the vector space.

For this study, the glove (6B,100d) pre-trained model was used. This version was trained on six billion tokens, with a vocabulary of 400,000 uncased words and 100-dimensional embeddings. The relatively small dimensionality ensured computational efficiency while preserving sufficient semantic richness for downstream comparison.

A custom procedure, implemented through a function was used to generate sentence-level embeddings by averaging the GloVe vectors of constituent words. Unknown tokens were ignored in the averaging process to prevent bias from unseen vocabulary. Each sentence was tokenised and its embedding was calculated as the mean of all known word vectors. Sentences were then written to a memory-mapped NumPy file.

3.2.4 Modelling

This study explores two distinct deep learning architectures for modelling user–item interactions based on review text: the DeepCoNN model and the prajjwal1/bert-tiny Transformer-based model. Both approaches aim to leverage textual review data to learn latent representations of users and items, but they differ fundamentally in their

architectural design and the mechanisms through which they capture semantic information. DeepCoNN employs parallel CNNs to extract features from review text and models user-item interactions through a shared factorisation layer, whereas the bert-tiny model utilises a compact Transformer encoder to generate contextualised embeddings from text. By comparing these two approaches, the study evaluates the effectiveness of convolutional and Transformer-based architectures in representing semantic and contextual patterns within review data for rating prediction.

3.2.4.1 DeepCoNN Model Architecture

The DeepCoNN model was employed to jointly learn representations of users and items from review text for rating prediction. DeepCoNN integrates the semantic information contained in textual reviews with CNNs to capture complex patterns in user behaviour and item properties (Zheng *et al.*, 2017). The model architecture comprises two parallel CNNs, one dedicated to users (denoted as $Net_{(u)}$) and the other to items ($Net_{(i)}$). Each network takes as input the corresponding set of reviews, represented as word embeddings, and outputs latent feature vectors that describe user preferences and item characteristics. These feature representations are combined in a shared layer that models their interactions to predict the user’s rating for an item.

Convolution and Pooling Layers

The CNN layers apply multiple convolutional filters to learn high-level textual features from the embedded review matrix. Each filter $K_j \in \mathbb{R}^{c \times t}$ operates on a window of t words to generate feature maps:

$$z_j = f(V_{u1:n} * K_j + b_j) \quad (3.2)$$

where b_j is a bias term and $f(x) = \max(0, x)$ is the Rectified Linear Unit (ReLU) activation function. Max-pooling is then applied to capture the most salient feature in each map:

$$o_j = \max\{z_1, z_2, \dots, z_{n-t+1}\} \quad (3.3)$$

The outputs from all filters form a feature vector $O = \{o_1, o_2, \dots, o_{n_1}\}$, which is then passed through a fully connected layer:

$$x_u = f(WO + g) \quad (3.4)$$

where W and g represent the weight matrix and bias vector respectively. This process produces the latent feature representation x_u for the user network. The same process is applied to the item network, resulting in feature representation y_i .

Shared Interaction Layer

To couple the two networks and enable their latent factors to interact, a shared layer is introduced on top. The user and item feature vectors are concatenated into a single vector $\hat{z} = (x_u, y_i)$, and their interactions are modelled using a Factorisation Machine (FM) to capture both first-order and second-order feature relationships:

$$J = \hat{w}_0 + \sum_{i=1}^{|\hat{z}|} \hat{w}_i \hat{z}_i + \sum_{i=1}^{|\hat{z}|} \sum_{j=i+1}^{|\hat{z}|} \langle \hat{v}_i, \hat{v}_j \rangle \hat{z}_i \hat{z}_j \quad (3.5)$$

Here, $\hat{w}_0$ is a global bias, $\hat{w}_i$ represents the importance of the i -th feature, and $\langle \hat{v}_i, \hat{v}_j \rangle$ denotes the dot product between the latent vectors corresponding to the i -th and j -th variables. This formulation allows the model to efficiently capture complex pairwise interactions between user and item features.

3.2.4.2 BERT Tiny Model Architecture

The prajjwal1/bert-tiny model represents a compact variant of the Bidirectional Encoder Representations from Transformers (BERT) architecture, developed to achieve a balance between representational capacity and computational efficiency. It originates from the work of [Turc et al. \(2019\)](#), which demonstrated that smaller Transformer-based models, when pre-trained on large-scale text corpora, can attain competitive performance on downstream natural language understanding tasks.

Model Overview

The bert-tiny configuration consists of two Transformer encoder layers ($L = 2$), a hidden size of 128 ($H = 128$), and two self-attention heads, amounting to approximately 4.4 million parameters([Turc et al., 2019](#)). Despite its compactness, it retains the bidirectional contextual learning capability that characterises larger BERT models.

The model was pre-trained using a masked language modelling (MLM) objective and next sentence prediction task on large general-domain corpora, including BookCorpus and English Wikipedia. This pre-training phase equips the model to capture syntactic and semantic linguistic dependencies, an capability it retains despite its comparatively reduced network depth and dimensionality.

Model Implementation

In this study, bert-tiny was used as the backbone of a modified DeepCoNN framework, to encode textual reviews for both users and items. The model was implemented using the Transformers library in PyTorch, allowing local loading of the pre-trained weights from prajjwal1/bert-tiny. The configuration specified a hidden size of 128, two hidden layers, two attention heads, an intermediate size of 512, and a dropout rate of 0.1.

Each review text was tokenised and initially represented using pre-trained 100-dimensional GloVe embeddings or Mini-LM sentence embeddings. These vectors were projected to match the model’s hidden size through a linear transformation layer. The transformed embeddings were passed to the Transformer encoder, where the self-attention mechanism captured token-level dependencies, producing contextualised word representations. To ensure accurate attention weighting, Boolean attention masks were applied to distinguish padded tokens from meaningful text.

The encoder’s output was aggregated through mean pooling across the sequence length, producing a fixed-dimensional embedding for each review. Optional freezing of the Transformer backbone prevented updates to the pre-trained parameters, preserving the linguistic knowledge learned during pre-training.

Integration into DeepCoNN Framework

Two independent bert-tiny encoders were used to represent users and items, following the dual-branch structure of DeepCoNN. Each encoder processed up to ten review documents per user or item ($rc = 10$), and the outputs were averaged to form consolidated user and item representations. These were concatenated and passed through a fully connected multilayer perceptron (MLP) with a ReLU activation function and dropout regularisation to predict the final rating.

Basis and Relevance

The bert-tiny model builds on the Transformer encoder proposed by Vaswani *et al.* (2017), which uses multi-head self-attention and position-wise feed-forward networks to model token interactions across sequences. Although compact, BERT-tiny preserves the essential inductive bias of attention-based architectures by contextualising each token relative to its surrounding words. Its design aligns with findings from Bhargava *et al.* (2021), who examined model generalisation across Natural Language Inference datasets and found that even smaller BERT-based models can learn transferable linguistic features when properly trained. Therefore, bert-tiny is particularly suitable for applications that require an efficient yet expressive representation of text, such as recommender systems processing extensive review corpora.

3.2.4.3 Experimental Setup

All experiments were conducted in Kaggle Notebooks using the accelerator environment. The platform provided access to two NVIDIA T4 GPUs for up to 30 hours per week, which supported the efficient execution of model training and evaluation. Python served as the primary programming language, and all experimental workflows were implemented within this environment.

A range of standard scientific and machine learning libraries were used. Model development relied on `PyTorch`, the `Transformers` library, and associated `HuggingFace` tooling. Data manipulation and numerical operations were performed with `pandas` and `numpy`, while `scikit-learn` supported preprocessing, model evaluation, and utility functions. Visualisations were produced using `matplotlib` and `seaborn`. Additional Python standard libraries, including `collections` and others required for data handling and experimentation, were also employed where appropriate.

3.2.5 Evaluation

The evaluation was designed to compare the performance across two methodological dimensions: the choice of textual representation and the choice of the backbone architecture. The first comparison involved static GloVe embeddings versus contextual embeddings from all-MiniLM-L6-v2. The second comparison evaluated CNN backbones

against Transformer backbones. Together, these dimensions yielded four experimental configurations: CNN with GloVe embeddings, CNN with Transformer embeddings, Transformer backbone with GloVe embeddings, and Transformer backbone with contextual embeddings.

The recommender system task was considered under both regression and classification frameworks, separate evaluation metrics were employed to quantify model performance appropriately.

Regression Evaluation

When the task was framed as a regression problem, the model’s predictions were treated as continuous numerical estimates of user ratings. Two primary metrics were applied: Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). The MSE quantifies the average of the squared differences between the predicted values $\hat{y}_i$ and the actual ratings y_i , defined mathematically as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (3.6)$$

where n denotes the total number of predictions. This measure penalises large deviations more heavily due to the squaring of errors, making it sensitive to outliers.

The RMSE provides a more interpretable form of this metric by expressing the error in the same units as the predicted ratings:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (3.7)$$

A lower RMSE value indicates closer alignment between predicted and actual ratings, implying better predictive accuracy. During training, the MSE was calculated for both the training and validation datasets at each epoch. The validation MSE was monitored to guide early stopping, ensuring that model learning ceased when performance on unseen data began to deteriorate. This practice prevented overfitting and promoted generalisation. The model parameters corresponding to the lowest validation MSE were saved as the optimal configuration.

To supplement the MSE-based evaluation, the Mean Absolute Error (MAE) was also calculated, defined as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (3.8)$$

Unlike MSE, the MAE weights all errors linearly and is therefore less sensitive to extreme deviations. Together, these measures provided a comprehensive view of model accuracy and stability across the dataset.

Classification Evaluation

In the classification framework, continuous rating values were discretised into categorical sentiment classes to assess model performance in a qualitative sense. The binning strategy followed the mapping:

$$r \in [1, 2] \rightarrow \text{Negative (class 0)}, \quad r = 3 \rightarrow \text{Neutral (class 1)}, \quad r \in [4, 5] \rightarrow \text{Positive (class 2)}$$

The resulting classification task involved three sentiment categories. To account for potential imbalance in class distributions, class weights were calculated as the inverse of class frequencies:

$$w_c = \frac{n_c + \epsilon}{N} \quad (3.9)$$

where w_c is the weight for class c , n_c the number of samples in class c , N the total number of samples, and ϵ a small constant to prevent division by zero. These weights were incorporated into the cross-entropy loss function to ensure balanced learning across all categories.

Classification performance was assessed using accuracy, precision, recall, and F1-score, which are defined as follows:

Accuracy measures the overall proportion of correct predictions:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.10)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives respectively.

Precision indicates the proportion of correctly predicted positive instances among all instances predicted as positive:

$$Precision = \frac{TP}{TP + FP} \quad (3.11)$$

Recall (or sensitivity) reflects the proportion of actual positive instances that were correctly identified:

$$Recall = \frac{TP}{TP + FN} \quad (3.12)$$

The F1-score, the harmonic mean of precision and recall, provides a balanced measure of both metrics:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3.13)$$

In addition to these numerical metrics, a confusion matrix was constructed to visualise prediction errors across the three sentiment categories. The confusion matrix M is defined as a $C \times C$ matrix where each entry $M_{i,j}$ represents the number of observations known to belong to class i but predicted as class j . Diagonal elements indicate correct classifications, while off-diagonal elements represent misclassifications.

The combination of quantitative and visual evaluations facilitated a thorough understanding of model strengths and weaknesses.

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents and interprets the experimental findings obtained from implementing the proposed models on the *Amazon All.Beauty* dataset. The analysis focuses on two main dimensions: (i) the effect of different text embedding techniques: GloVe word embeddings and Transformer-based sentence embeddings generated using the *all-MiniLM-L6-v2* model and (ii) the effect of model architecture, comparing the DeepCoNN model with the *prajjwal1/bert-tiny* Transformer model.

Two predictive tasks were undertaken. The first was a regression task, which evaluated model performance using RMSE and MAE on the test set. The second was a three class classification task, which predicted discrete sentiment labels (Negative, Neutral, Positive) and was evaluated via accuracy and detailed per-class metrics. The chapter first presents the quantitative results for each model and embedding configuration, followed by a joint discussion of performance trends, task-specific patterns, and comparisons with prior work.

4.2 Experimental Runs and Quantitative Results

Two embedding techniques were explored: MiniLM-L6-v2 sentence embeddings and GloVe word embeddings. Each embedding type was paired with two model architectures: DeepCoNN and *prajjwal1/bert-tiny*. The following subsections present the regression and classification results for each configuration. Early stopping was applied in all runs based on validation performance.

4.2.1 MiniLM-L6-v2 Embeddings with DeepCoNN Model Architecture

4.2.1.1 Regression Results

Table 4.1: Regression performance for MiniLM-L6-v2 embeddings with DeepCoNN model.

Epoch	Train RMSE	Validation RMSE
0	1.1192	1.0328
1	1.0700	1.0298
2	1.0603	1.0217
3	1.0508	1.0055
4	1.2527	1.0386
Best validation RMSE: 1.0055 (Epoch 3)		
Train RMSE: 0.9974 Valid RMSE: 1.0055 Valid MAE: 0.7243		
Test RMSE: 1.0022 Test MAE: 0.7225		

Table 4.1 presents the regression results across epochs. The model achieved the best validation RMSE of 1.0055 at epoch 3. Final evaluation on the test set yielded an RMSE of 1.0022 and an MAE of 0.7225, indicating stable performance across train, validation, and test splits.

4.2.1.2 Classification Results

Table 4.2: Classification performance across epochs for MiniLM-L6-v2 embeddings with DeepCoNN model.

Epoch	Train Loss	Validation Accuracy
0	0.8201	0.7351
1	0.7924	0.7101
2	0.7816	0.7004
3	0.7733	0.7213
Best validation accuracy: 0.7351 (Epoch 0)		

Table 4.2 reports validation performance for the classification task. The model achieved its highest validation accuracy of 0.7351 at epoch 0, after which early stopping

4.2 Experimental Runs and Quantitative Results

Table 4.3: Classification report for MiniLM-L6-v2 embeddings with DeepCoNN model.

Class	Precision	Recall	F1-Score	Support
Negative	0.6781	0.6512	0.6644	14409
Neutral	0.1983	0.5017	0.2842	5657
Positive	0.9368	0.7853	0.8544	50015
Accuracy			0.7348	70081
Macro Average	0.6044	0.6461	0.6010	70081
Weighted Average	0.8240	0.7348	0.7693	70081

was triggered. The detailed classification report for the test set is shown in Table 4.3. The overall accuracy was 0.7348, with the model performing particularly well on the positive class (F1-score of 0.8544). The neutral class, by contrast, exhibited lower precision and recall, indicating greater prediction difficulty.

4.2.2 GloVe Embeddings with DeepCoNN Model Architecture

4.2.2.1 Regression Results

Table 4.4: Regression performance for GloVe embeddings with DeepCoNN model.

Epoch	Train RMSE	Validation RMSE
0	1.2164	1.1521
1	1.1680	1.1254
2	1.1720	1.1180
3	1.2731	1.1587
4	1.2763	1.1488
Best validation RMSE: 1.1180 (Epoch 2)		
Test RMSE: 1.1110 Test MAE: 0.8253		

Table 4.4 summarises the regression performance. The best validation RMSE of 1.1180 was achieved at epoch 2. On the test set, the model obtained an RMSE of 1.1110 and a MAE of 0.8253, both higher than those of the corresponding MiniLM-based configuration.

4.2 Experimental Runs and Quantitative Results

Table 4.5: Classification performance across epochs for GloVe embeddings with DeepCoNN model.

Epoch	Train Loss	Validation Accuracy
0	0.8759	0.7322
1	0.8517	0.6722
2	0.8440	0.6799
3	0.8392	0.6992
Best validation accuracy: 0.7322 (Epoch 0)		

Table 4.6: Classification report for GloVe embeddings with DeepCoNN model.

Class	Precision	Recall	F1-Score	Support
Negative	0.5419	0.7799	0.6395	14409
Neutral	0.1933	0.2533	0.2193	5657
Positive	0.9192	0.7706	0.8383	50015
Accuracy			0.7307	70081
Macro Average	0.5515	0.6013	0.5657	70081
Weighted Average	0.7830	0.7307	0.7475	70081

4.2.2.2 Classification Results

Table 4.5 reports classification performance across epochs. The best validation accuracy of 0.7322 was obtained at epoch 0. The classification report for the test set is shown in Table 4.6. Overall accuracy was 0.7307, with strong performance on the positive class (F1-score of 0.8383), but relatively weak performance for the neutral class.

4.2.3 MiniLM-L6-v2 Embeddings with *prajjwal1/bert-tiny* Model Architecture

4.2.3.1 Regression Results

Table 4.7 summarises regression performance for MiniLM embeddings paired with the *prajjwal1/bert-tiny* model. The best validation RMSE of 1.0101 was achieved at epoch 2, and the test set yielded an RMSE of 1.0013 and a MAE of 0.6673.

4.2 Experimental Runs and Quantitative Results

Table 4.7: Regression performance for MiniLM-L6-v2 embeddings with *prajjwal1/bert-tiny* model.

Epoch	Train RMSE	Validation RMSE
0	1.0782	1.0358
1	1.0418	1.0208
2	1.0270	1.0101
3	1.0164	1.0185
4	1.0085	1.0143
Best validation RMSE: 1.0101 (Epoch 2)		
Test RMSE: 1.0013 Test MAE: 0.6673		

4.2.3.2 Classification Results

Table 4.8 presents the classification results across epochs. The highest validation accuracy of 0.8234 was attained at epoch 4, with a corresponding test accuracy of 0.8230. The detailed classification report is given in Table 4.9. The model achieved very strong performance on the positive class (F1-score of 0.9030), but substantially weaker recall for the neutral class.

Table 4.8: Classification accuracy across epochs for MiniLM-L6-v2 embeddings with *prajjwal1/bert-tiny* model.

Epoch	Train Accuracy	Validation Accuracy
0	0.8061	0.8154
1	0.8145	0.8197
2	0.8177	0.8205
3	0.8199	0.8205
4	0.8211	0.8234
Best validation accuracy: 0.8234 (Epoch 4)		
Test accuracy: 0.8230		

4.2 Experimental Runs and Quantitative Results

Table 4.9: Classification report for MiniLM-L6-v2 embeddings with *prajjwal1/bert-tiny* model.

Class	Precision	Recall	F1-Score	Support
Negative	0.6970	0.7089	0.7029	14409
Neutral	0.3511	0.0896	0.1428	5657
Positive	0.8698	0.9388	0.9030	50015
Accuracy			0.8230	70081
Macro Average	0.6393	0.5791	0.5829	70081
Weighted Average	0.7924	0.8230	0.8005	70081

4.2.4 GloVe Embeddings with *prajjwal1/bert-tiny* Model Architecture

4.2.4.1 Regression Results

Table 4.10 presents regression performance for GloVe embeddings combined with *prajjwal1/bert-tiny*. The lowest validation RMSE of 1.1000 was achieved at epoch 4. On the test set, the model yielded an RMSE of 1.0893 and an MAE of 0.7394.

Table 4.10: Regression performance for GloVe embeddings with *prajjwal1/bert-tiny* model.

Epoch	Train RMSE	Validation RMSE	Validation MAE
0	1.1771	1.1395	0.7795
1	1.1434	1.1283	0.7608
2	1.1294	1.1322	0.8329
3	1.1197	1.1062	0.7852
4	1.1118	1.1000	0.7463
Best validation RMSE: 1.1000 (Epoch 4)			
Test RMSE: 1.0893 Test MAE: 0.7394			

4.2.4.2 Classification Results

Table 4.11 summarises the classification performance across epochs. The best validation accuracy of 0.6951 occurred at epoch 1, and the test accuracy was 0.6958. The classification report in Table 4.12 shows strong performance on the positive class but

4.3 Overall Performance Trends

Table 4.11: Classification accuracy across epochs for GloVe embeddings with *prajjwal1/bert-tiny* model.

Epoch	Train Accuracy	Validation Accuracy
0	0.6438	0.6346
1	0.6838	0.6951
2	0.6758	0.6812
3	0.6538	0.6648
Best validation accuracy: 0.6951 (Epoch 1)		
Test accuracy: 0.6958		

Table 4.12: Classification report for GloVe embeddings with *prajjwal1/bert-tiny* model.

Class	Precision	Recall	F1-Score	Support
Negative	0.6260	0.6097	0.6177	14409
Neutral	0.1523	0.3946	0.2198	5657
Positive	0.9118	0.7547	0.8258	50015
Accuracy			0.6958	70081
Macro Average	0.5634	0.5863	0.5545	70081
Weighted Average	0.7917	0.6958	0.7341	70081

weaker precision and recall for the neutral class, reflecting the class imbalance and difficulty of modelling neutrality.

4.3 Overall Performance Trends

Table 4.13 summarises the test-set performance metrics for all four experimental configurations and highlights the combined effects of embedding choice and model architecture.

Two clear patterns emerge. First, configurations using MiniLM-L6-v2 sentence embeddings consistently outperform those using GloVe embeddings across all metrics. Second, the *prajjwal1/bert-tiny* architecture combined with MiniLM-L6-v2 achieves the strongest classification performance and the lowest MAE, whereas the MiniLM + DeepCoNN configuration obtains the best RMSE.

Table 4.13: Summary of experimental results for all embedding and model configurations (test set).

Embedding	Model	RMSE	MAE	Accuracy
MiniLM-L6-v2	<i>prajjwal1/bert-tiny</i>	1.0013	0.6673	0.8230
MiniLM-L6-v2	DeepCoNN	1.0022	0.7225	0.7348
GloVe	<i>prajjwal1/bert-tiny</i>	1.0893	0.7394	0.6958
GloVe	DeepCoNN	1.1110	0.8253	0.7307

4.3.1 Effects of Embedding Techniques

The comparison between embedding techniques reveals substantial performance differences. MiniLM-L6-v2 sentence embeddings provide contextualised representations at the sentence level, allowing the model to capture semantic nuance and word meaning in context. This capability is particularly important for product reviews, where the same word can carry different sentiment depending on surrounding text.

By contrast, GloVe offers static, context-independent word embeddings. Each token is represented by a fixed vector regardless of its usage context, which limits the model’s ability to differentiate subtle shifts in tone or mixed sentiment in user-generated reviews.

The superiority of MiniLM embeddings is most pronounced in classification. The best MiniLM configuration (MiniLM + *prajjwal1/bert-tiny*) achieves 82.30% accuracy, compared to 73.07% for the best GloVe-based configuration (GloVe + DeepCoNN). This corresponds to a relative improvement of approximately 12.6%, underscoring the importance of contextualised representations for sentiment classification and rating prediction.

4.3.2 Effects of Model Architecture

The results also reveal important architectural effects. The *prajjwal1/bert-tiny* model achieves the highest classification accuracy and best MAE when combined with MiniLM embeddings. Its Transformer-based architecture appears particularly well-suited for exploiting contextual information to make stable classification decisions.

DeepCoNN, by contrast, exhibits competitive and in some cases superior performance in regression, achieving the lowest RMSE of 1.0022 when paired with MiniLM

embeddings. Its dual CNN structure, which separately models user and item representations, may help to reduce large prediction errors, benefiting RMSE.

An important trade-off thus emerges: DeepCoNN provides slightly better RMSE, while *prajjwal1/bert-tiny* offers more consistent predictions overall, as reflected in lower MAE and higher classification accuracy. This suggests that the choice of architecture should be guided by whether the application prioritises reducing large rating errors or maximising consistent sentiment categorisation.

4.4 Task-Specific Performance Patterns

4.4.1 Classification Performance and Confusion Matrix Analysis

Figure 4.1 summarises test accuracy across configurations. The MiniLM + *prajjwal1/bert-tiny* model clearly dominates, with an absolute gain of about 9 percentage points over the next-best configuration.

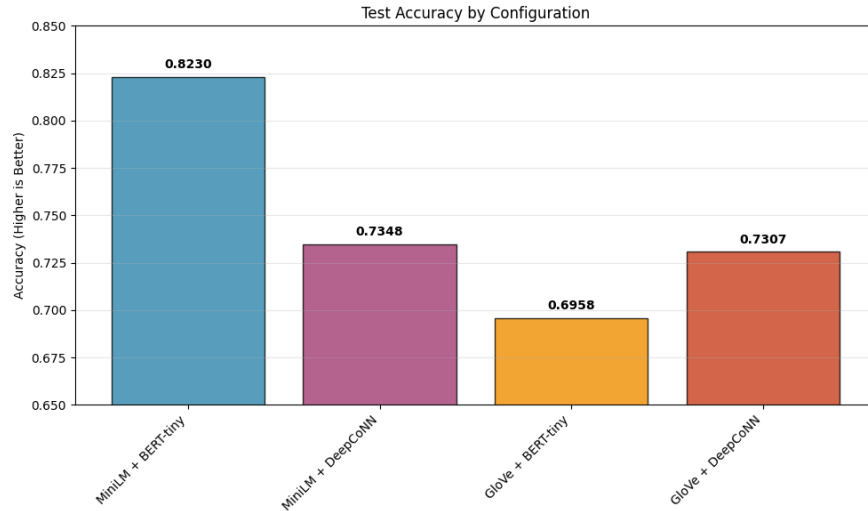


Figure 4.1: Accuracy comparison across different configurations.

While accuracy provides a useful global ranking, it does not describe how errors are distributed across classes. To address this, row-normalised confusion matrices were examined for each configuration. These matrices reveal systematic differences in how models treat negative, neutral, and positive sentiment.

4.4.1.1 GloVe Embeddings + DeepCoNN Architecture

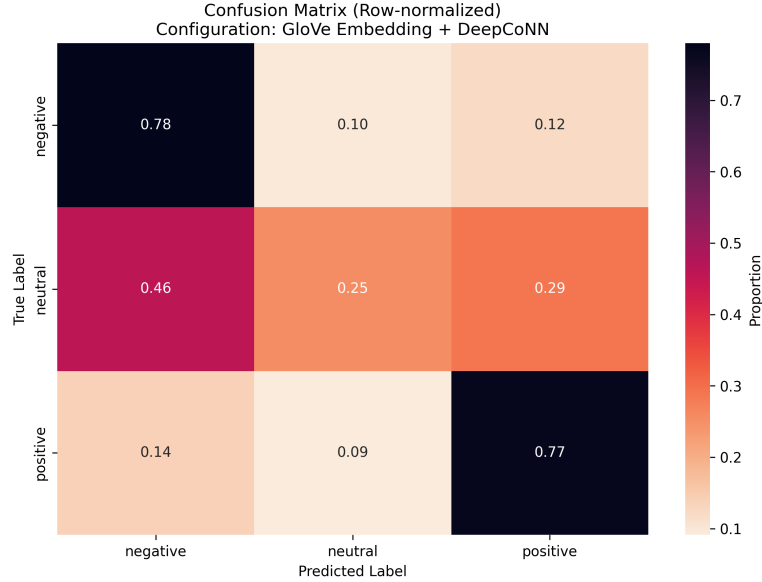


Figure 4.2: Row-normalised confusion matrix for the GloVe + DeepCoNN configuration.

The GloVe + DeepCoNN configuration (Figure 4.2) performs strongly on the sentiment extremes. It correctly identifies 78% of negative reviews and 77% of positive reviews, indicating that clearly polarised language is well captured. However, neutral reviews are challenging: only 25% of neutral instances are correctly classified, with most being mislabelled as negative or positive. The model therefore maintains robust decision boundaries for negative and positive classes, but fails to carve out a well-defined region for neutrality in its learnt representation space.

4.4.1.2 GloVe Embeddings + BERT-Tiny Architecture

When GloVe embeddings are paired with the *prajjwal1/bert-tiny* architecture (Figure 4.3), classifier behaviour becomes more internally inconsistent. Positive recall remains reasonably high at 75%, but negative recall drops to 61%. Neutral recall improves to 39%, yet neutral instances are still widely dispersed across all classes. At the same time, a notable fraction of strongly polarised reviews are softened into the neutral category (26% of negative and 17% of positive reviews misclassified as neutral), suggesting

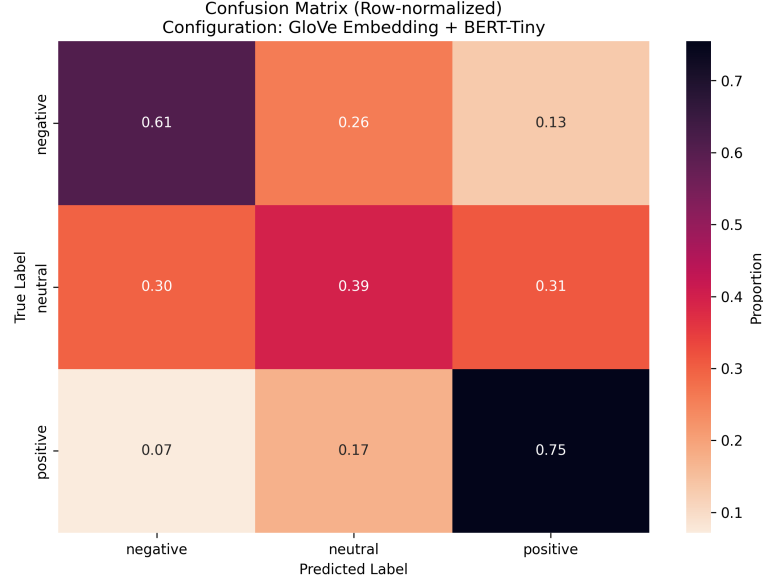


Figure 4.3: Row-normalised confusion matrix for the GloVe + BERT-Tiny configuration.

that the model struggles to maintain clear polarity boundaries when relying on static embeddings within a compact Transformer.

4.4.1.3 MiniLM Embeddings + DeepCoNN Architecture

Replacing GloVe with MiniLM embeddings within DeepCoNN yields a more balanced classifier (Figure 4.4). Neutral recall doubles to 50%, indicating that the contextual information provided by MiniLM substantially improves recognition of low-polarity and objective language. This gain comes with a trade-off in the negative class: negative recall declines to 65%, and 26% of true negative reviews are re-assigned to the neutral category. Positive recall remains high at 79%, the best among the first three configurations. Overall, this configuration is particularly suitable when accurate identification of neutral feedback is important, and a modest reduction in negative sensitivity is acceptable.

4.4.1.4 MiniLM Embeddings + BERT-Tiny Architecture

The MiniLM + *prajjwal1/bert-tiny* configuration (Figure 4.5) shows a highly polarised and atypical profile. The model attains exceptional positive recall of 94%, the highest

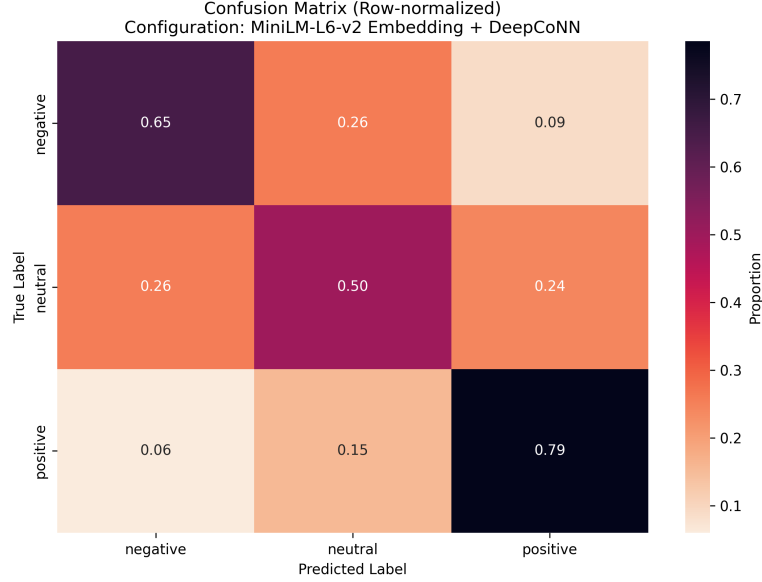


Figure 4.4: Row-normalised confusion matrix for the MiniLM + DeepCoNN configuration.

recall for any class across all configurations. However, neutral recall collapses to 9%, with 57% of neutral reviews misclassified as positive. Additionally, 26% of negative reviews are erroneously classified as positive, leading to problematic polarity inversions where clearly negative sentiment is interpreted as positive. This configuration therefore excels in identifying strongly positive sentiment but lacks reliable boundaries for neutrality and mismanages a significant subset of negative reviews.

4.4.1.5 Comparative Insights and Structural Challenges

Taken together, the confusion matrices reveal three overarching themes. First, the choice of embedding emerges as the primary driver of sentiment nuance. GloVe-based models struggle to define the neutral class, repeatedly collapsing neutral instances into the polar categories. In contrast, MiniLM-based models, particularly the MiniLM + DeepCoNN configuration, demonstrate a substantially improved ability to distinguish neutrality. This indicates that contextualised embeddings are critical for effectively modelling intermediate sentiment.

Second, the model architecture significantly shapes how these embeddings are exploited. DeepCoNN acts as a stable translator of embedding quality; when paired with

4.4 Task-Specific Performance Patterns

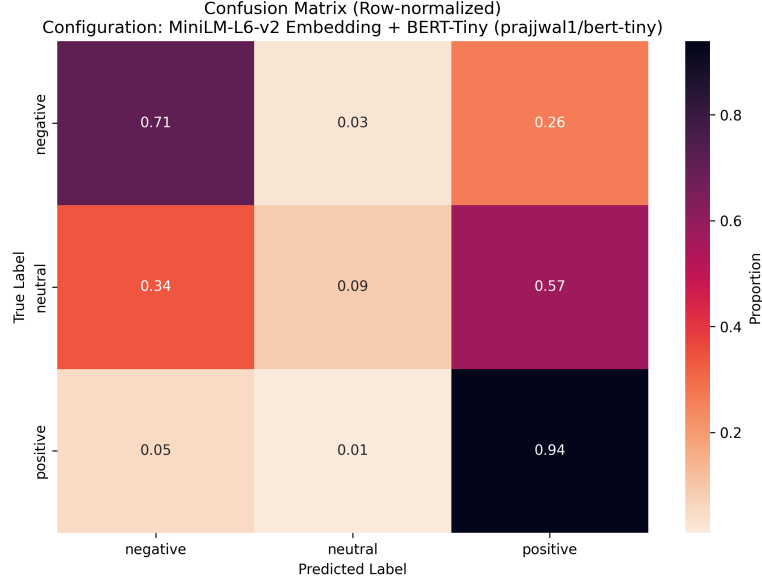


Figure 4.5: Row-normalised confusion matrix for the MiniLM + BERT-Tiny configuration.

GloVe, it functions as a strong polarity detector, whereas with MiniLM, it evolves into a more balanced classifier. The *Prajjwal1/bert-tiny* model, by contrast, shows greater sensitivity to its configuration. With GloVe, it produces mixed class boundaries, and with MiniLM, it exhibits a tendency to over-commit to the positive class. This suggests that further tuning or additional model capacity may be requisite for this architecture to achieve balanced three-class sentiment modelling.

Finally, the analysis confirms that the neutral class is structurally difficult to model. No configuration achieves high recall for all three classes simultaneously. A recurring trade-off is observed: enhancing neutral detection tends to compress the representation space available for negative sentiment, as seen in the MiniLM + DeepCoNN model. Conversely, amplifying positive detection undermines the classification of both neutral and negative instances, a pattern evident in the MiniLM + *bert-tiny* results. This persistent challenge points to an inherent difficulty in three-way sentiment formulations, where neutral language is often linguistically and semantically ambiguous.

These patterns closely align with prior findings such as those of (Chu *et al.*, 2021), who report that BERT-based sentiment classification with rating binning reaches a test accuracy plateau around 83% and struggles particularly with the neutral or mid-range

rating band. Their eventual shift from classification to regression reflects the same structural difficulty in modelling neutrality that is observed here.

4.4.2 Regression Performance

For the regression task, the performance differences between configurations are more nuanced than in classification. MiniLM + DeepCoNN achieves the best RMSE (1.0022), while MiniLM + *prajjwal1/bert-tiny* achieves the best MAE (0.6673). This discrepancy suggests that DeepCoNN is slightly better at avoiding large errors, which RMSE penalises quadratically, whereas *bert-tiny* yields more uniformly distributed errors, which reduces MAE.

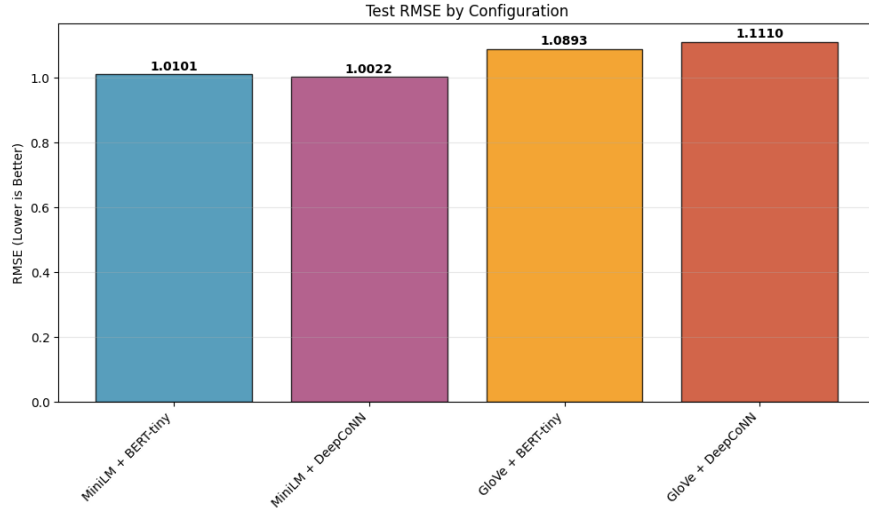


Figure 4.6: RMSE comparison across configurations.

As illustrated in Figures 4.6 and 4.7, RMSE varies from 1.0022 to 1.1110, corresponding to roughly a 10.9% spread between best and worst performers, while MAE ranges from 0.6673 to 0.8253, representing about a 23.7% improvement from the weakest to the strongest configuration. In both metrics, MiniLM-based configurations consistently outperform GloVe-based ones, reinforcing the central role of contextual embeddings.

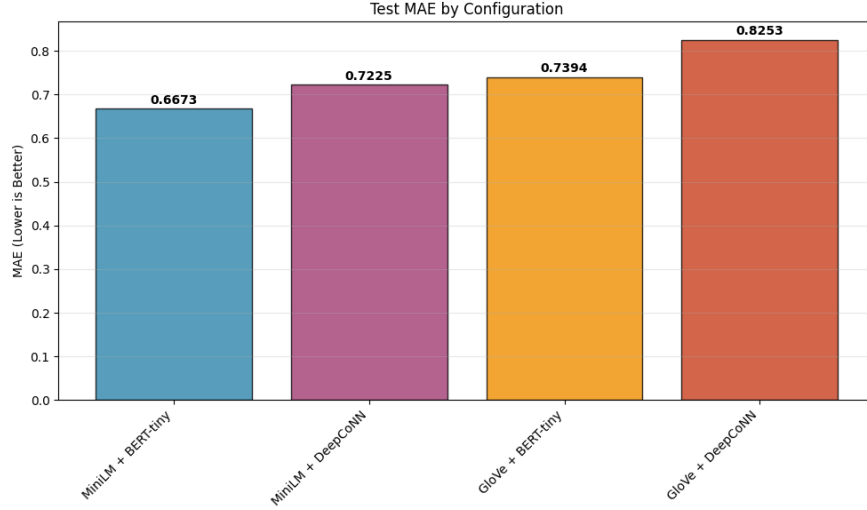


Figure 4.7: MAE comparison across configurations.

4.5 Comparison with Previous Studies

To contextualise these findings, it is useful to compare them with prior work on rating prediction and review-based recommendation. [Gheewala *et al.* \(2024\)](#) evaluated a range of deep learning and Transformer-based architectures on the Amazon Electronics and Amazon Movies & TVs datasets. Their regression results for CNN-, LSTM-, and attention-based models with GLoVe and BERT embeddings are summarised in Table 4.14, while Transformer models such as BERT, RoBERTa, XLNet, GPT-2, MPNet and ELECTRA are reported in Table 4.15.

Their results show that models using static GLoVe embeddings consistently produce higher RMSE and MAE than those using contextual BERT embeddings. For instance, CNN + FM with GLoVe yields RMSE values of 1.326 and 1.329 on the two datasets, whereas CNN + FM with BERT reduces these to 1.078 and 1.051. Transformer-based models operating directly on contextual representations perform even better: XLNet + FM achieves RMSE values of 0.978 and 0.976, with MAE as low as 0.699.

The best configuration in the present study, MiniLM-L6-v2 embeddings combined with *prajjwal1/bert-tiny*, attains a test RMSE of 1.0013 and MAE of 0.6673 on the Amazon All.Beauty dataset. Although these RMSE values are slightly higher than those reported for larger Transformer models such as XLNet, they are competitive

4.5 Comparison with Previous Studies

Table 4.14: RMSE and MAE results from Table 7 for Amazon Electronics and Amazon Movies & TVs dataset from the study by [Gheewala *et al.* \(2024\)](#).

Model	Embed	Amazon Electronics		Amazon Movies & TVs	
		RMSE	MAE	RMSE	MAE
CNN + FM	GLoVe	1.326	0.978	1.329	0.978
CNN + FM	BERT	1.078	0.831	1.051	0.810
LSTM + FM	GLoVe	1.288	1.059	1.486	1.257
LSTM + FM	BERT	1.037	0.773	1.029	0.782
CNN-Attn + FM	GLoVe	1.329	0.957	1.302	0.960
CNN-Attn + FM	BERT	1.072	0.827	1.038	0.804
LSTM-Attn + FM	GLoVe	1.431	1.253	1.463	1.282
LSTM-Attn + FM	BERT	1.025	0.755	1.049	0.792
CNN + NCF	GLoVe	1.349	1.040	1.381	1.018
CNN + NCF	BERT	1.171	0.914	1.115	0.925
LSTM + NCF	GLoVe	1.309	1.072	1.463	1.282
LSTM + NCF	BERT	1.138	0.935	1.193	1.004
LSTM-Attn + NCF	GLoVe	1.270	1.047	1.462	1.255
LSTM-Attn + NCF	BERT	1.049	0.792	1.113	0.879

given the different domain and the more lightweight architecture used. The MiniLM + DeepCoNN configuration, with RMSE of 1.0022, also compares favourably with BERT + FM and GPT-2 + FM in Table 4.15.

Crucially, both this study and prior work converge on the same conclusion: contextual Transformer-based embeddings substantially improve rating prediction compared to static GLoVe embeddings. The present results extend this observation to a different Amazon domain (All Beauty) and demonstrate that compact Transformers such as *prajjwal1/bert-tiny* can deliver competitive performance when paired with high-quality sentence embeddings.

At the same time, the persistent difficulty of modelling the neutral class aligns with [Chu *et al.* \(2021\)](#), who report that three-way sentiment classification with rating-based bins plateaus around 83% accuracy and particularly struggles to model mid-range ratings. Both studies highlight that review ratings behave more like a continuous scale

4.6 Summary of Key Findings

Table 4.15: RMSE and MAE results from Table 8 for Amazon Electronics and Amazon Movies & TVs dataset from the study by [Gheewala *et al.* \(2024\)](#).

Transformer Model	Amazon Electronics		Amazon Movies & TVs	
	RMSE	MAE	RMSE	MAE
BERT + FM	1.008	0.735	0.985	0.726
RoBERTa + FM	1.206	0.931	0.988	0.728
XLNet + FM	0.978	0.699	0.976	0.735
GPT-2 + FM	1.002	0.753	0.983	0.726
MPNet + FM	1.013	0.744	0.986	0.731
ELECTRA + FM	1.012	0.745	0.991	0.723
BERT + NCF	1.169	0.872	1.102	0.873
RoBERTa + NCF	1.212	0.943	1.113	0.886
XLNet + NCF	1.050	0.793	1.095	0.874
GPT-2 + NCF	1.037	0.794	1.097	0.883
MPNet + NCF	1.199	0.944	1.107	0.871
ELECTRA + NCF	1.064	0.799	1.094	0.868

than discrete sentiment categories, and that imposing a three-class structure introduces structural tensions that cannot be fully resolved by simply scaling model size or swapping architectures.

4.6 Summary of Key Findings

A synthesis of the quantitative results, confusion matrix analyses, and comparisons with prior work yields three key conclusions. First, the use of contextual sentence embeddings is demonstrated to be essential for fine-grained sentiment modelling. The MiniLM-L6-v2 model consistently surpassed the GloVe baseline across both regression and classification tasks, achieving a relative improvement in accuracy of more than 12% and substantially enhancing the model’s capacity to recognise the nuanced neutral sentiment class.

Second, the choice of model architecture induces a distinct trade-off between error magnitude and prediction consistency. The MiniLM + DeepCoNN configuration achieved the most favourable RMSE, indicating a superior ability to control large pre-

diction errors. In contrast, MiniLM + *bert-tiny* yielded the best MAE and accuracy, suggesting more uniform prediction quality across the dataset. Consequently, the selection of the most appropriate model configuration is contingent upon the application’s specific priority: whether to minimise severe rating deviations or to maximise overall classification reliability.

Finally, the modelling of neutrality persists as a structural challenge within three-class sentiment frameworks. Across all tested architectures and embeddings, the neutral class was consistently the most difficult to identify accurately. Furthermore, performance improvements in one region of the sentiment spectrum, such as positive or negative, frequently resulted in degradations elsewhere. This recurring trade-off mirrors patterns documented in the existing literature and reinforces the perspective that sentiment, as expressed in ratings, is inherently continuous. Any discrete classification scheme applied to this continuous space inevitably imposes representational compromises.

Together, these findings support the use of contextual Transformer-based embeddings and lightweight Transformer architectures for review-based rating prediction, while also highlighting the intrinsic limits of three-class sentiment formulations in domains where ratings reflect fine-grained and often ambiguous user judgements.

Chapter 5

Conclusions

This chapter concludes the thesis by summarising the main findings, reflecting on the contributions to academic knowledge and practice, acknowledging limitations, and outlining directions for future research. The work investigated how different combinations of text embeddings and model architectures influence the performance of review-based recommender systems, with a particular focus on three-class sentiment classification and rating prediction from unstructured user feedback.

5.1 Summary of Key Findings

The empirical results demonstrate that embedding choice is the dominant factor shaping model behaviour. Across all architectures and tasks, MiniLM-L6-v2 sentence embeddings consistently outperformed static GloVe embeddings in terms of RMSE, MAE, and classification accuracy. MiniLM-based configurations showed a markedly improved capacity to capture contextual nuance in beauty product reviews, especially in distinguishing neutral or low-polarity language from clearly positive or negative sentiment. GloVe-based models, by contrast, struggled to represent neutrality and tended to project ambiguous reviews towards the sentiment extremes.

Model architecture exerted a secondary, but still important, influence. When combined with MiniLM-L6-v2 embeddings, the *prajjwal1/bert-tiny* transformer achieved the highest overall classification accuracy (82.30%) and the lowest MAE (0.6673), indicating more consistent predictions across the rating spectrum. DeepCoNN, however, achieved the best RMSE (1.0022) when paired with MiniLM-L6-v2, suggesting that its

dual CNN design is particularly effective at avoiding large individual errors in rating prediction. These results point to a clear trade-off between architectures: the compact transformer offered more stable average performance, while DeepCoNN reduced extreme deviations.

A detailed analysis of row-normalised confusion matrices revealed a persistent structural challenge in modelling the neutral class. No configuration achieved simultaneously high recall for negative, neutral, and positive sentiment. Instead, improvements in neutral recall were systematically associated with reductions in negative recall, and vice versa. GloVe + DeepCoNN behaved as a strong polarity detector but captured neutrality poorly; MiniLM + DeepCoNN improved neutral recall substantially but became more conservative in labelling negative sentiment. MiniLM + *prajjwal1/bert-tiny* delivered exceptional performance on positive reviews but collapsed neutrality into positivity and misclassified a non-trivial proportion of negative reviews as positive. This pattern confirms that balanced three-class sentiment modelling remains difficult, particularly in the presence of class imbalance.

Comparison with related work showed that these findings are consistent with earlier studies. Prior research has reported similar difficulties in modelling mid-range ratings and neutral sentiment and has often concluded that ratings behave more like a continuous scale than a set of discrete categories. The regression results obtained here are broadly comparable with those reported for larger transformer models in the literature, despite the use of smaller architectures and more constrained computational resources. This suggests that lightweight transformer backbones, when combined with high-quality contextual embeddings, can deliver competitive predictive accuracy in real-world settings.

5.2 Addressing the Research Questions

The study was guided by three research questions. The findings allow the following conclusions to be drawn.

RQ1: How do transformer-based embeddings compare to static word embeddings in representing unstructured user feedback?

The results provide clear evidence that transformer-based sentence embeddings offer substantial advantages over static word embeddings for representing unstructured user feedback. Across all configurations, MiniLM-L6-v2 embeddings produced lower RMSE and MAE values and higher classification accuracy than GloVe. The confusion matrix analyses further showed that MiniLM-based models were better able to distinguish neutral reviews from polarised sentiment and to capture nuanced linguistic cues typical of beauty product discourse. RQ1 is therefore well addressed: transformer-based contextual embeddings outperform static word embeddings in both predictive performance and representational adequacy for this task.

RQ2: Does a transformer-based backbone architecture outperform a CNN baseline when provided with identical input features?

The comparison between *prajjwal1/bert-tiny* and DeepCoNN, both consuming identical embeddings, indicates that the transformer backbone does not uniformly dominate the CNN baseline under the experimental conditions used. When paired with MiniLM-L6-v2 embeddings, the transformer achieved higher classification accuracy and lower MAE, whereas DeepCoNN achieved slightly better RMSE and, in some configurations, more balanced class-specific performance. When paired with GloVe embeddings, DeepCoNN generally outperformed the transformer. These findings suggest that architectural superiority is conditional on the task and metric considered, and that the compact transformer used here may require more careful tuning to fully realise its potential. RQ2 is therefore only partially addressed: the transformer backbone offers advantages in certain respects but does not consistently outperform the CNN baseline under the constraints imposed in this study.

RQ3: What is the combined effect of employing transformer embeddings together with a transformer backbone on recommendation accuracy and efficiency?

The combined use of MiniLM-L6-v2 embeddings and the *prajjwal1/bert-tiny* backbone led to the best overall classification performance and competitive regression metrics, demonstrating that this configuration can achieve high accuracy with relatively modest

computational requirements. In light of the resource constraints and dataset subsampling imposed by the Kaggle environment, these results show that a compact transformer backbone coupled with efficient sentence embeddings can deliver performance approaching that of larger models reported in previous work, while remaining feasible for practitioners without access to extensive GPU resources. RQ3 is therefore addressed by showing that the combined use of transformer embeddings and a lightweight transformer backbone yields a favourable accuracy–efficiency trade-off in the context of review-based recommendation.

5.3 Contributions to Knowledge and Practice

5.3.1 Academic Contributions

The thesis makes several contributions to academic understanding of text-based recommender systems and sentiment modelling:

- It provides a controlled comparison of static word embeddings and transformer-based sentence embeddings within identical downstream architectures, demonstrating empirically that contextual embeddings are crucial for capturing neutral and nuanced sentiment in review data.
- It offers a systematic architectural comparison between a CNN-based recommendation model (DeepCoNN) and a compact transformer backbone, using matched input features. This highlights how different architectures trade off between aggregate accuracy, error dispersion, and class-wise performance.
- It presents a detailed confusion-matrix-based analysis of three-class sentiment classification, showing that balanced modelling of negative, neutral, and positive sentiment is structurally difficult, particularly under class imbalance. The analysis clarifies the nature of the trade-offs between neutral and negative recall across embedding–architecture combinations.
- It situates the findings within the wider literature on sentiment-based rating prediction, demonstrating that compact transformers paired with strong sentence embeddings can produce results comparable with much larger models, thereby

extending prior work on the accuracy–complexity trade-off in recommender systems.

5.3.2 Practical Implications

The study also provides guidance for practitioners designing review-based recommender systems:

- The results suggest that embedding choice should be prioritised over marginal architectural variants when resources are limited. Adopting high-quality contextual sentence embeddings, such as MiniLM-L6-v2, is likely to yield greater gains than relying on static word embeddings.
- For applications constrained by computational resources, the combination of efficient sentence embeddings with a compact transformer backbone can deliver strong performance without the cost of training or deploying large-scale models. This makes transformer-based techniques more accessible in production environments with limited GPU capacity.
- The persistent difficulties in modelling neutral sentiment indicate that practitioners should treat neutral predictions with caution, especially when decisions or recommendations depend on correctly identifying mid-range user satisfaction. In some settings, regression or ordinal approaches may provide a more faithful representation of the underlying rating continuum than discrete three-class classification.

5.4 Limitations

Limitations of the study should be acknowledged. First, the experiments were conducted on the AllBeauty subset of the Amazon review data, which is highly imbalanced, with a strong dominance of five-star ratings. This imbalance affected both regression and classification tasks. In classification, the scarcity of mid-range ratings contributed to weak performance on the neutral class and encouraged models, particularly MiniLM + *prajjwal1/bert-tiny*, to develop a bias towards positive predictions. As a result, the findings may not generalise directly to domains with more evenly distributed ratings or different sentiment dynamics.

Second, the computational environment imposed significant constraints. The work was carried out in Kaggle notebooks, where GPU usage is time-limited. These constraints necessitated the use of smaller datasets, reduced opportunities for extensive hyperparameter searches, and restricted the choice of architectures to relatively compact transformer models. Larger transformer backbones and more extensive fine-tuning schedules could not be explored within these limits.

Third, the study focused on a narrow set of embedding–architecture combinations: GloVe versus MiniLM-L6-v2, and DeepCoNN versus *prajjwal1/bert-tiny*. While this design allowed for clear comparisons, it does not exhaust the space of potentially relevant models, particularly more recent or larger transformer variants.

Together, these limitations delimit the scope of the conclusions and indicate that the results should be interpreted as evidence under a specific domain, dataset, and resource setting rather than as universally applicable benchmarks.

5.5 Directions for Future Work

Building on these limitations and findings, several avenues for future research are apparent. First, there is clear scope for more extensive optimisation of the transformer backbone. Systematic fine-tuning of *prajjwal1/bert-tiny* (for instance, through more extensive hyperparameter exploration, alternative learning-rate schedules, and regularisation strategies) could test whether the observed biases and weaknesses are primarily configuration-related. Extending the analysis to larger transformer architectures, such as BERT-base, RoBERTa, or XLNet, would further clarify how model capacity interacts with sentence embeddings in both classification and regression settings.

Second, future work should broaden the empirical basis by incorporating additional datasets and domains. Evaluating the proposed configurations on other Amazon categories or different platforms would test whether the observed behaviours, particularly the difficulty of modelling neutrality, are specific to beauty products or generalise across contexts. Including datasets with more balanced rating distributions would also help disentangle the effects of class imbalance from those of model design.

Third, several modelling refinements merit investigation. Techniques such as data resampling, class-weighted loss functions, or focal loss could be employed to mitigate

class imbalance and improve neutral-class performance. Alternative output formulations, such as ordinal regression or joint classification–regression objectives, may better capture the continuous nature of ratings and reduce boundary tensions between sentiment categories.

Finally, future research could examine the integration of richer user and item side information and explore end-to-end architectures in which embeddings and backbones are trained jointly rather than in a fixed-feature setting. Such work would help determine whether the advantages of contextual embeddings observed here persist or are amplified when models are allowed to adapt all components to the target domain.

In conclusion, this work demonstrates that transformer-based embeddings and compact transformer architectures can materially improve the modelling of unstructured review text in recommender systems, while also highlighting persistent challenges in representing neutral sentiment and balancing performance under realistic resource constraints. These findings provide a foundation for more fine-grained, and generalisable approaches to sentiment-aware recommendation in future work.

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